

Registered Report

Action Anticipation Based on an Agent's Epistemic State in Toddlers and Adults

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The registered Stage 1 manuscript unchanged from the point of IPA can be found on the Open Science Framework: <https://osf.io/7ubkw>. We have no known conflicts of interests to disclose. Data are available at the following URL: <https://osf.io/jmuvd/>. The analytic code necessary to reproduce the preregistered analyses presented in this paper is available at <https://github.com/manybabies/mb2-analysis>. The materials necessary to attempt to replicate the findings presented here are publicly accessible at the following URL: <https://osf.io/pd35f/files>.

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Abstract

Do toddlers and adults engage in spontaneous Theory of Mind? This multi-lab collaboration examined whether 18- to 27-month-olds' and adults' anticipatory looks distinguish between two basic forms of epistemic states: knowledge and ignorance. In adults ($n = 703$, 68% female), we found clear evidence that they do: they showed simple goal-based action anticipation in pilot studies and differentiated between knowledge and ignorance conditions in the main study. In toddlers ($n = 521$, 49% female), results were less conclusive. While demonstrating goal-based action anticipation in pilot studies, they did not differentiate between knowledge and ignorance as predicted. Future research can explore adults' sensitivity to more complex epistemic states like true/false beliefs and clarify whether toddlers' results reflect competence or performance limitations.

Keywords: anticipatory looking; spontaneous Theory of Mind; replication

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Action Anticipation Based on an Agent's Epistemic State in Toddlers and Adults

The capacity to represent epistemic states, known as Theory of Mind (ToM) or mentalizing, plays a central role in human cognition (Dennett, 1989; Frith & Frith, 2006; Premack & Woodruff, 1978). Although ToM has been under intense scrutiny in the past decades, its nature and ontogeny are still the subjects of much controversy. At the heart of these debates are questions about the reliability of the paradigms used to measure ToM (Baillargeon et al., 2018; Poulin-Dubois et al., 2018), including, among others, anticipatory looking (AL) paradigms. To address this issue, in a collaborative long-term project we assess the robustness of infants' and adults' tendency to spontaneously take into account different kinds of epistemic states — what they perceive, know, think, or believe — when predicting others' behaviors. This paper reports the first foundational step of this project, which focuses on the most basic epistemic state ascription: the capacity to distinguish between knowledgeable and ignorant individuals. Simple forms of knowledge attribution (such as tracking what other individuals have seen or experienced) are typically assumed to develop early and to operate spontaneously throughout the lifespan (Liszkowski et al., 2007; Luo & Baillargeon, 2007; O'Neill, 1996; Phillips et al., 2021). Thus, evaluating whether ToM measures are sensitive to the knowledge-ignorance distinction is a crucial test case to assess their robustness. The present paper investigates this question in an AL paradigm including 18-27-month-old infants and adults.

In the following sections we first establish the background and scientific context of this study, namely the reliability and replicability of spontaneous ToM measures. We then introduce a novel way to approach these issues: a large-scale collaborative project targeting the replicability of ToM findings. Finally, we outline the rationale of the present study which uses an AL paradigm to test whether infants and adults distinguish between two basic forms of an agent's epistemic state: knowledge and ignorance.

Spontaneous Theory of Mind tasks

Humans are proficient at interpreting and predicting others' intentional actions. Adults as well as infants expect agents to act persistently towards the goal they pursue (Csibra & Gergely, 2007; Gergely & Csibra, 2003; Gergely et al., 1995, Woodward & Sommerville, 2000), and they anticipate others' actions based on their goals even before goals are achieved - that is, humans engage in goal-based action anticipation (for review, see Elsner & Adam, 2021; but see Ganglmayer et al., 2019). To predict others' actions, however, it is essential to consider their epistemic state: what they perceive, know, or believe. A number of seminal studies using non-verbal spontaneous measures have suggested that infants, toddlers, older children, and adults show action anticipation and action understanding not only based on other agents' goals (what they want) but also on the basis of their epistemic status (what they perceive, know, or believe). These studies suggest that from infancy onwards, humans spontaneously engage in ToM or mentalizing. For example, studies using the violation of expectation paradigm have demonstrated that infants look longer in response to events in which an agent acts in ways that are incompatible with their (true or false) beliefs, compared to events in which they act in belief-congruent ways (Onishi & Baillargeon, 2005; Surian et al., 2007; Träuble et al., 2010). Other studies have employed more interactive tasks requiring the child to play, communicate, or cooperate with experimenters and, for example, give an experimenter one of several objects as a function of their epistemic status. Such studies have shown that toddlers spontaneously adjust their behavior to the experimenter's beliefs (Buttelmann et al., 2009; Király et al., 2018; Knudsen & Liszkowski, 2012; Southgate et al., 2010).

The largest body of evidence for spontaneous ToM comes from studies using AL tasks. In such tasks, participants see an agent who acts in pursuit of some goal (typically, to collect a certain object) and has either a true or a false belief (for example, regarding the location of the

target object). A number of studies have shown that infants, toddlers, older children, neurotypical adults, and even non-human primates anticipate (as indicated by looks to the location in question) that an agent will go where it (truly or falsely) believes the object to be, irrespective of the actual location of the object (Gliga et al., 2014; Grosse Wiesmann et al., 2017; Hayashi et al., 2020; Kano et al., 2019; Krupenye et al., 2016; Meristo et al., 2012; Schneider et al., 2012; Schneider et al., 2013; Senju et al., 2009; Senju et al., 2010; Senju et al., 2011; Surian & Franchin, 2020; Thoermer et al., 2012). These studies have revealed converging evidence for spontaneous ToM across the human lifespan and even in other primate species.

Across the different measures, the majority of early works on spontaneous ToM in infants and toddlers have reported positive results in the second year of life, and a few studies even within the first year (Kovács et al., 2010; Luo, 2011; Southgate & Vernetti, 2014), yielding a rich body of coherent and convergent evidence (for reviews see e.g., Barone et al., 2019; Kamps et al., 2020; Scott & Baillargeon, 2017). This growing body of literature has led to a theoretical transformation of the field. In particular, findings with young infants have paved the way for novel accounts of the development and cognitive foundations of ToM. The previous consensus was that full-fledged ToM emerges only at around age 4, potentially as the result of developing executive functions, complex language skills and other factors (e.g., Perner, 1991; Wellman & Cross, 2001). In contrast, the newer accounts proposed that some basic forms of ToM may be phylogenetically more ancient and may develop much earlier in ontogeny (e.g., Baillargeon et al., 2010; Carruthers, 2013; Kovács, 2016; Leslie, 2005).

Recently, however, a number of studies have raised uncertainty regarding the empirical foundations of the early-emergence theories, as we review below. In the following sections, we present an overview of the current empirical picture of early understanding of epistemic states

and then introduce ManyBabies2 (MB2), a large-scale collaborative project exploring the replicability of ToM in infancy, of which the current study constitutes the first step.

Replicability of Spontaneous Theory of Mind Tasks

A number of failures to replicate findings from spontaneous ToM tasks have recently been published with infants, toddlers, and adults (e.g., Burnside et al., 2018; Dörrenberg et al., 2018; Grosse Wiesmann et al., 2017; Grosse Wiesmann et al., 2018; Kamps et al., 2021; Kulke, von Duhn, et al., 2018; Kulke & Hinrichs, 2021; Kulke, Johannsen, & Rakoczy, 2019; Kulke & Rakoczy, 2017, 2019; Kulke, Reiß, et al., 2018; Kulke, Wübker, & Rakoczy, 2019; Powell et al., 2018; Priewasser et al., 2020; Priewasser et al., 2018; Schuwerk et al., 2018; for overviews, see Barone et al., 2019; Kulke & Rakoczy, 2018). Besides conceptual replications, many of these studies involve more direct replication attempts with the original stimuli and procedures. One of these was a two-lab replication attempt of one of the most influential AL studies (Southgate et al., 2007). This failure to replicate is especially notable not only because of the influence of the original finding of the field, but also because of the large sample size and the involvement of some of the original authors (Kamps et al., 2021). Additional unpublished replication failures have also been reported. Kulke and Rakoczy (2018) examined 65 published and non-published studies including 36 AL studies (replications of Low & Watts, 2013; Schneider et al., 2012; Southgate et al., 2007; Surian & Geraci, 2012), as well as studies using other paradigms, and classified them as a successful, partial, or non-replication, depending on whether all, some, or none of the original main effects were found. Although no formal analysis of effect size was carried out, overall, non-replications and partial replications outnumbered successful replications, regardless of the method used.

In addition to the failure to replicate spontaneous anticipation of agents' behaviors based on their beliefs, many of the replication studies revealed an even more fundamental problem of spontaneous AL procedures: a failure to adequately anticipate an agent's action in the absence of a belief. That is, researchers did not find evidence for spontaneous anticipation of agents' behaviors based on their goals, even in the initial familiarization trials of the experiments (e.g., Kamps et al., 2020; Kulke, Reiß, et al., 2018; Schuwerk et al., 2018). The familiarization trials are designed to convey the goal of the agent, as well as the general timing and structure of events, to set up participants' expectations in the test trials where the agent's epistemic state is then manipulated. Typically, the last familiarization trial can also be used to probe participants' spontaneous action anticipation; and test trials can only be meaningfully interpreted if there is evidence of above-chance anticipation in the familiarization trials. In several AL studies many participants had to be excluded from the main analyses for failing to demonstrate robust action anticipation during the familiarization trials (e.g., Kamps et al., 2020; Kulke, Reiß, et al., 2018; Schuwerk et al., 2018; Southgate et al., 2007). This raises the possibility that these AL paradigms may not be suitable for reliably eliciting spontaneous action prediction in the first place (for discussion see Baillargeon et al., 2018; Kamps et al., 2021). In light of the complex and mixed state of the evidence, researchers called for a systematic, large-scale, multi-lab study to stringently test the robustness, reliability, and replicability of AL and other spontaneous measures of ToM.

General Rationale of MB2

To this end, MB2 was established as an international consortium dedicated to investigating infants' and toddlers' ToM skills. The main aim is to test the replicability and thus reliability of findings from spontaneous ToM tasks. In the long-term, MB2 will build on the

initial findings and the aim will be extended to include testing the validity of these experimental designs and addressing theoretical accounts of spontaneous ToM. MB2 operates under the general umbrella of ManyBabies (MB), a large-scale international research consortium founded with the aim of probing the reliability of central findings from infancy research. In particular, MB projects bring together large and theoretically diverse groups of researchers to tackle pressing questions of infant cognitive development, by collaboratively designing and implementing methodologies and pre-registered analysis plans (Frank et al., 2017). The MB2 consortium involves authors of original studies as well as authors of both successful and failed replication studies, and researchers from very different theoretical backgrounds. It thus presents a case of true “adversarial collaboration” (Mellers et al., 2001).

Rationale of the Present Study

Based on both theoretical and practical considerations, the current paper presents the first foundational step in MB2, focusing on AL measures. It investigates whether toddlers and adults anticipate (in their looking behavior) how other agents will act based on their goals (i.e., what they want) and epistemic status (i.e., what they know or do not know). From a practical perspective, we focus on AL since it is a child-friendly and widely used method that is also suitable for humans across the lifespan and even other species. Additionally, as AL is screen-based and standardizable, identical stimuli can be presented in different labs. From a theoretical perspective, given the mixed findings with AL tasks reviewed in the previous section, we take a systematic and bottom-up approach.

First, we probe whether AL measures are suitable for measuring spontaneous goal-directed action anticipation. With the aim of improving the low overall rates of anticipatory looks in recent studies, we designed new, engaging stimuli to test whether these are successful in

eliciting spontaneous action anticipation. Second, in case reliably elicited action anticipation can be found: we probe whether toddlers and adults take into account the agent's epistemic status in their spontaneous goal-based action anticipation. That is, do they track whether the agent saw or did not see a crucial event, and therefore whether this agent does or does not know something? In the current study we focus on the most basic form of tracking the epistemic status of agents: considering whether they had access to relevant information, and whether they are thus *knowledgeable* or *ignorant*. We reasoned that only after establishing whether a context can elicit spontaneous tracking of an agent's epistemic status in a more basic sense (i.e., the agent's knowledge vs. ignorance) is it eventually meaningful to ask whether this context also elicits more complex epistemic state tracking (i.e., the agent's beliefs).

Answering these first two questions in the present study will allow us, in the long run, to address a third set of questions in subsequent studies, probing the nature of the representations and cognitive mechanisms involved in infant ToM. Do toddlers and adults engage in full-fledged belief-ascription in their spontaneous goal-based action anticipation? What *kind* of epistemic states do toddlers and adults spontaneously attribute to others in their action anticipation (e.g., Horschler et al., 2020; Phillips et al., 2021)? Do the results that prove replicable really assess ToM, or can they be interpreted in alternative ways such as behavioral rules, associations, or simple perceptual preferences (see, e.g., Heyes, 2014; Perner & Ruffman, 2005)? The present study lays the foundation for investigating these questions.

Regarding the knowledge-ignorance distinction, many accounts in developmental and comparative ToM research have argued for the ontogenetic and evolutionary primacy of representing *what* agents witness and represent, relative to more sophisticated ways of representing *how* agents represent (and potentially mis-represent) objects and situations (e.g., Apperly & Butterfill, 2009; Flavell, 1988; Kaminski et al., 2008; Martin & Santos, 2016; Perner,

1991; Phillips et al., 2021). For example, it is often assumed that young children and non-human primates may be capable of so-called “Level I perspective-taking” (understanding *who* sees *what*) but only human children from around age 4 may finally develop capacities for “Level II perspective-taking” (understanding *how* a given situation may appear to different agents; Flavell et al., 1981). Empirically, many studies using verbal and/or interactive measures have indicated that children may engage in knowledge-ignorance and related distinctions before they engage in more complex forms of meta-representation (e.g., Flavell et al., 1981; Hogrefe et al., 1986; Moll & Tomasello, 2006; O’Neill, 1996; Buttelmann & Kovács, 2019; Buttelmann et al., 2015; Kampis et al., 2020; though for some findings indicating Level II perspective-taking at an early age see Forgács et al., 2019; Scott & Baillargeon, 2009; Scott et al., 2015), and that non-human primates seem to master knowledge-ignorance tasks while not demonstrating any more complex, meta-representational form of ToM (e.g., Hare et al., 2001; Kaminski et al., 2008; Karg et al., 2015). The knowledge-ignorance distinction thus appears to be an ideal candidate for assessing epistemic status-based action anticipation in a wide range of populations.

To date, however, no study has probed whether or how children’s (and adults’) spontaneous action anticipation, as indicated by AL, is sensitive to ascriptions of knowledge vs. ignorance. Most studies that have addressed ToM with AL measures have targeted the more sophisticated true/false belief contrast. As reviewed above, the results of those studies yield a mixed picture regarding replicability of the findings. It has been argued that tasks that reliably replicate are ones which can be solved with the more basic knowledge-ignorance distinction, whereas tasks that do not replicate require more sophisticated belief-ascription (Powell et al., 2018)¹, suggesting that only some findings might not be replicable. Based on these

¹ For example, some studies have found partial replication results, with patterns of the following kind: participants showed systematic anticipation (or appropriate interactive responses) in true belief trials but showed looking (or interactive responses) at chance level in the false belief trials (e.g., Dörrenberg et al., 2019; Kulke, Reiß,

considerations, the present study tests whether toddlers and adults engage in knowledge- and ignorance-based AL to probe the most basic form of spontaneous, epistemic state-based action anticipation.

Design and Predictions of the Present Study

The current study presented 18- to 27-month-old toddlers and adults with animated scenarios while measuring their gaze behavior. Testing adults (and not just toddlers) is crucial to address debates about the validity and interpretation of AL measures of ToM throughout the lifespan (e.g., Schneider et al., 2017). Following the structure of previous AL paradigms, participants were first familiarized to an agent repeatedly approaching a target (familiarization trials). AL was measured during the familiarization trials to probe whether participants understood the agent's goal and spontaneously anticipate their actions. Subsequently, during the test trials the agent's visual access was manipulated, leading them to be either knowledgeable or ignorant about the location of the target. Participants' AL was measured during the test trials to determine whether or not they take into account the agent's epistemic access and adjust their action anticipation accordingly. Participants' looking patterns were recorded using either lab-based corneal reflection eye-tracking or online recording of gaze patterns. We chose to provide the online testing option to increase the flexibility for data collection given the disruption caused by the COVID-19 pandemic. This option also provided the opportunity to potentially compare in-lab and online testing procedures (Sheskin et al., 2020; see Section S4 of Supplemental Material).

Novel animated stimuli were collectively developed within the MB2 consortium on the basis of previous work (e.g., Clements & Perner, 1994) and based on input from collaborators

et al., 2018; Powell et al., 2018). Such a pattern remains ambiguous since it may merely reflect a knowledge-ignorance distinction.

with experience with both successful and failed replication studies (e.g., Grosse Wiesmann et al., 2017; Surian & Geraci, 2012). These animated 3D scenes featured a dynamic interaction aimed to optimally engage participants' attention: a chasing scenario involving two agents, a *chaser* and a *chasee* (see Figures 1 and 2). As part of the chase, the chasee enters from the top of an upside-down Y-shaped tunnel with two boxes at its exits. The tunnel is opaque so participants cannot see the chasee after it enters the tunnel, but can hear noises that indicate movement. The chasee eventually exits from one of the arms of the Y, and goes into the box on that side. The chaser observes the chasee exit the tunnel and go into a box, and then follows it through the tunnel. During familiarization trials, the chaser always exits the tunnel on the same side as the chasee, and approaches the box where the chasee is currently located. Thus, if participants engage in spontaneous action anticipation during familiarization trials, they should reliably anticipate during the period when the chaser is in the tunnel that it will emerge at the exit that leads to the box containing the chasee.

During test trials, the chasee always first hides in one of the boxes but shortly thereafter leaves its initial hiding place and hides in the box at the other tunnel exit. Critically, the chaser either does (*knowledge* condition) or does not (*ignorance* condition) have epistemic access to the chasee's location. During knowledge trials, the chaser observes all movements of the chasee. During ignorance trials, the chaser observes the chasee enter the tunnel, but then leaves and only returns after the chasee is hidden inside the second box. The event sequences in the two conditions are thus identical with the only difference between conditions pertaining to what the chaser has or has not seen. They were designed in this way with the long-term aim to implement, in a minimal contrast design, more complex conditions of false/true belief contrasts with the very same event sequences (true belief conditions will then be identical to the knowledge conditions

here, but in false belief conditions the chaser witnesses the chasee's placement in the first box, but then fails to witness the re-location)².

Participants' AL (their gaze pattern indicating where they expect the chaser to appear) will be assessed during the anticipatory period - that is, the period during which the chaser is going through the tunnel and is not visible. There will be two main dependent measures: first looks, and a differential looking score (DLS). The first look measure will be binary, indicating which of the two tunnel exits participants fixate first: the exit where the chasee is actually hiding, or the other exit. DLS is a measure of the proportion of time spent looking at the correct tunnel exit during the entire anticipatory period.

In two pilot studies (see Methods section), we addressed the foundational question of the current study: whether these stimuli reveal spontaneous goal-directed action anticipation as measured by AL in the above-described familiarization trials (i.e., without a change of location by the chasee or manipulation of the chaser's epistemic state). We found that our paradigm indeed elicited action anticipation and exclusion rates due to lack of anticipation were significantly lower relative to previous (original and replication) AL studies. Both toddlers and adults showed reliable anticipation of the chaser's exit at the chasee's location, indicating that in contrast with many previous AL studies the current paradigm successfully elicits spontaneous goal-based action anticipation. Based on these pilot data we concluded that the paradigm is

² There is thus a certain asymmetry with regard to the interpretation and the consequences of potentially positive and negative results of the present knowledge-ignorance contrast: in the case of positive results, we can conclude that participants spontaneously engage in basic epistemic state ascription and can move on to test, with the minimal contrast comparison of knowledge-ignorance vs. false belief-true belief, whether this extends to more complex forms of epistemic state attribution. In the case of negative results, though, we cannot draw firm conclusions to the effect that participants do not engage in spontaneous epistemic state ascription. More caution is in order since the present knowledge-ignorance contrast has been designed in order to be comparable to future belief contrasts rather than to be the simplest implementation possible. Simpler implementations would then need to be devised that involve fewer steps (i.e. the chasee just goes to one location and this is or is not witnessed by the chasee).

suitable for examining the second and critical question: whether toddlers and adults, in their spontaneous goal-based action anticipation, take into account the agent's epistemic state.

We predict that if participants track the chaser's perceptual access and resulting epistemic state (knowledge/ignorance) and anticipate their actions accordingly, they should look more in anticipation to the exit at the chasee's location than the other exit in the knowledge condition, but should not do so (or to a lesser degree; see below) in the ignorance condition. We anticipate three potential factors that could influence participant's gaze patterns: Keeping track of the chaser's epistemic status in the ignorance condition might either lead to no expectations as to where the chaser will look (resulting in chance level looking between the two exits) or (if participants follow an "ignorance leads to mistakes"-rule, see e.g., Ruffman, 1996) to an expectation that the chaser will go to the wrong location (longer looking to the exit with the empty box; e.g., Fabricius et al., 2010). Either way, participants may still show a 'pull of the real' even in the ignorance condition, i.e., reveal a default tendency to look to the side where the chasee is located. But if they truly keep track of the epistemic status of the chaser (knowledge vs. ignorance), they should show this tendency to look to the side where the chasee really is in the ignorance condition to a lesser degree than in the knowledge condition.

In sum, the research questions of the present study are the following: First, can we observe in a large sample that toddlers and adults robustly anticipate agents' actions based on their goals in this paradigm, as they did in our pilot study? Second, can we find evidence that they take into account the agent's epistemic access (knowledge vs. ignorance) and adjust their action anticipation accordingly? In addressing these questions, the present study will significantly contribute to our knowledge on spontaneous ToM. It will inform us whether the present paradigm and stimuli can elicit spontaneous goal-based and mental-state-based action anticipation in adults and toddlers, based on a large sample of 1224 participants in total from 47 labs. In the long run,

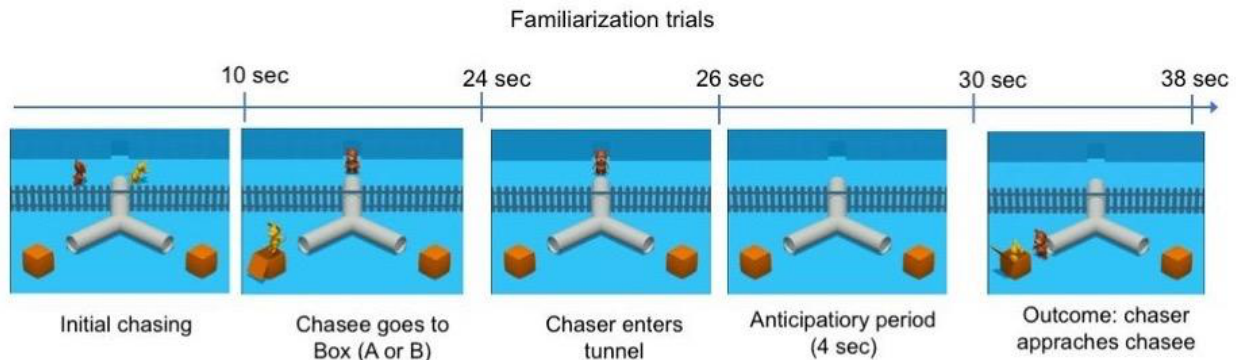
the present study will lay the foundation for future work to address broader questions of what *kind* of epistemic states toddlers and adults spontaneously attribute to others in their action anticipation and what cognitive mechanisms allow them to do so.

Methods

All materials, and the collected de-identified data, are provided on the Open Science Framework (OSF; <https://osf.io/jmuvd/>). All analysis scripts, including the pilot data analysis and simulations for the design analysis, can be found on GitHub (<https://github.com/manybabies/mb2-analysis>). We report how we determined our sample size and we report all data exclusions, all manipulations, and all measures in the study. Additional methodological details can be found in the Supplemental Material.

Stimuli

Figures 1 and 2 provide an overview of the paradigm. For the stimuli, 3D animations were created depicting a chasing scenario between two agents (chaser and chasee) who start in the upper part of the scene. At the very top of the scene a door leads to outside the visible scene. Below this area, a horizontal fence separates the space, and thus the lower part of the space can be reached by the Y-shaped tunnel only. Additional information on the general scene setup, events, and timings in the familiarization and the test trials, as well as trial randomization can be found in the Supplemental Material.

Figure 1*Timeline of the familiarization trials*

Note. The last picture illustrates the reunion of the two agents, after the chaser left the tunnel, approached the closed box (with chasee still inside) and knocked on it.

Familiarization Trials

All participants viewed four familiarization trials (for an overview of key events see Figure 1). During the familiarization trials, after a brief chasing introduction, the chasee enters an upside-down Y-shaped tunnel with a box at both of its exits. The chasee then leaves the tunnel through one of the exits and hides in the box on the corresponding side. Subsequently, the chaser enters the tunnel (to follow the chasee), and participants' AL to the tunnel exits is measured before the chaser exits on the side the chasee is hiding, as an index of their goal-based action anticipation. In these familiarization trials, if participants engage in spontaneous action anticipation, they should reliably anticipate that the chaser should emerge at the tunnel exit that leads to the box where the chasee is. After leaving the tunnel, the chaser approaches the box in which the chasee is hiding and knocks on it. Then, the chasee jumps out of the box and the two briefly interact.

Familiarization Phase Pilot Studies. In a pilot study with 18- to 27-month-olds ($n = 65$) and adults ($n = 42$), seven labs used in-lab corneal reflection eye-tracking to collect data on gaze behavior in the familiarization phase using eight trials. A key desideratum of our paradigm was that it should produce sufficient AL, as a low rate of AL in previous studies had led to high exclusion rates. The goals of the pilot study were to 1) estimate the level of correct goal-based action predictions in the familiarization phase, 2) determine the optimal number of familiarization trials, 3) check for issues with perceptual properties of stimuli (e.g., distracting visual salencies), and 4) test the general procedure including preprocessing and analyzing raw gaze data from different eye-tracking systems. We found that the familiarization stimuli elicited a relatively high proportion of goal-directed action anticipations, but we were concerned about the effects of some minor properties of the stimulus (in particular, a small rectangular window in the tunnel tube that allowed participants to see the agents at one point on their path to the tunnel exits).

In a second pilot study with 18- to 27-month-olds ($n = 12$, three participating labs), slight changes of stimulus features (the removal of the window in the tube; temporal changes of auditory anticipation cue) did not cause major changes in the AL rates.

Sixty-eight percent of toddlers' first looks in the first pilot, 69% of toddlers' first looks in the second pilot, and 69% of adults' first looks were toward the correct area of interest (AOI) during the anticipatory period. The average proportion of looking towards the correct AOI during the anticipatory period was 70.7% ($CI_{95\%} = 67.6\% - 73.8\%$) in toddlers in the first pilot, 70.5% ($CI_{95\%} = 62.8\% - 78.2\%$) in the second pilot for toddlers, and 75.3% ($CI_{95\%} = 71.0\% - 79.5\%$) in adults. In Bayesian analyses, we found strong evidence that toddlers and adults looked more towards the target than towards the distractor during the anticipation period. Based on conceptual and practical methodological considerations while also considering previous studies, we decided to include four trials in the final experiment. The pilot data results of the toddlers supported this

decision insofar as we observed a looking bias towards the correct location already in trials 1-4, without additional benefit of trials 5-8.

Further, prototypical analysis pipelines were established for combining raw gaze data from different eye-trackers. In short, we developed a way to resample gaze data from different eye-trackers to be at a common Hz rate and to define proportionally correct AOIs for different screen dimensions with the goal to merge all raw data into one data set for inferential statistics. The established analysis procedure is described further in the Data Preprocessing section below. In sum, we concluded that this paradigm sufficiently elicits goal-directed action predictions, an important prerequisite for drawing any conclusion on AL behavior in the test trials of this study. A detailed description of the two pilot studies can be found in the Supplemental Material.

Test Trials

All participants saw two test trials, one knowledge and one ignorance trial. However, in line with common practice in ToM studies, the main comparison concerned the first test trial between-participants to avoid potential carryover effects. In addition, in exploratory analyses, we assessed whether results remain the same if both trials were taken into account and whether gaze patterns differed between the two trials (see Exploratory Analyses). If the results remained largely unchanged across the two trials, it might suggest that future studies could increase power by including multiple test trials.

In the test trials, the chasee first hid in one of the boxes, but shortly thereafter the chasee left this box and hid in the second box, at the other tunnel exit. Critically, the chaser either witnessed (knowledge condition) or did not witness (ignorance condition) from which tunnel exit the chasee exited and thus where the chasee was currently hiding (for an overview, see Figure 2). In the knowledge trial, the chaser observed all movements of the chasee. The chaser leaves for a brief period of time after the chasee exited the tunnel, but returns before the chasee changes location from

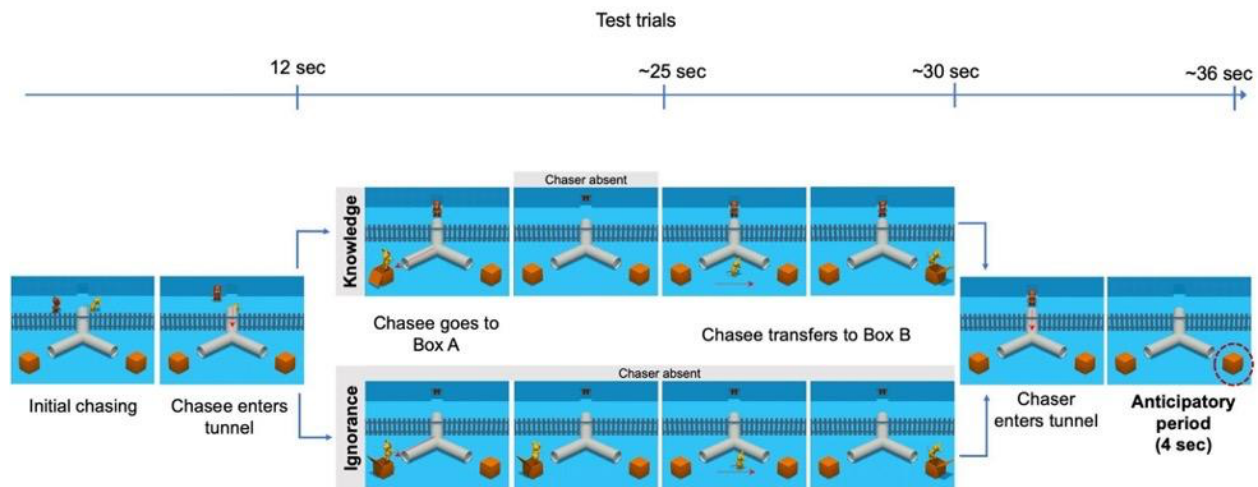
one box to the other. Therefore, no events took place in the chaser's absence. In the ignorance trial, the chaser saw the chatee enter the tunnel, but then left. Therefore, the chaser did not see the chatee entering either box and only returned once the chatee was already hidden in the final location. Finally, the chaser entered the tunnel but did not appear in either exit. Rather, the scene "froze" for four seconds and participants' AL was measured. Thus, the knowledge and ignorance conditions were matched for the chaser leaving for a period of time, but they differed in whether they warranted the chaser's epistemic access to the location of the chatee. No outcome was shown in either test trial.

When designing the knowledge and ignorance conditions, we tried to keep all events and their timings parallel, except for the crucial manipulation. We showed the same events in both conditions. Where possible, all events also had the same duration. In the case of the chaser's absence in the knowledge condition, there were two main options, both with inevitable trade-offs. First, we could have increased the duration of the chaser's absence in the knowledge condition to equate the duration of the chaser's absence in both conditions. Yet, this would have potentially disrupted the flow of events, such as keeping track of the chatee's actions and the general scene dynamics, since nothing would have happened for a substantial amount of time. Second, the chaser could have been absent for a shorter time in the knowledge than in the ignorance condition, in which case the flow of events – the chatee's actions and the general scene dynamics – would remain natural. We chose the second option because we reasoned that the artificial break in the knowledge condition could have disrupted the participant's tracking of the chaser's epistemic state, thus being a confound that would have been more detrimental than the difference in the duration of absence. Further, the current contrast had the advantage that the chatee's sequence and timing of actions were identical in both conditions, thus minimizing the difference between conditions. Finally, with the current design, the duration of the chaser's absence is

closely matched in the later planned false belief - true belief contrast, because in the future false belief condition, the chaser has to be absent for fewer events (because the chaser witnesses the first hiding events after the chasee reappeared at the other side of the tunnel).

Figure 2

Schematic overview of stimuli and conditions of the test trials



Note. After the familiarization phase, participants knew about the agent's goal (chaser wants to find chasee), perceptual access (chaser can see what happens on the other side of the fence), and situational constraints (boxes can be reached by walking through the forking tunnel). In the knowledge condition, the chaser witnessed the chasee walking through the tunnel and jumping in and out of the first box. While the chasee was in the box, the chaser briefly left the scene through the door in the back and returned shortly after. Subsequently, the chaser watched the chasee jumping out of the box again and hiding in the second box. In the ignorance condition, the chaser turned around and stood on the other side of the door in the back of the scene, thus unable to witness any of the chasee's actions. The chaser then returned and entered the tunnel to look for the chasee. During the test phase (4 seconds still frame), AL towards the end of the tunnels was measured.

Trial Randomization

We varied the starting location of the chasee (left or right half of the upper part of the scene) and the box the chasee ended up (left or right box) in both familiarization and test trials. The presentation of the familiarization trials was counterbalanced in two pseudo-randomized orders (LRRL and RLLR). Each lab signed up for one or two sets of 16-trial-combinations, for each of their tested age groups.

Lab Participation Details

Time-Frame

The contributing labs started data collection as soon as they were able to once our Registered Report received an in-principle acceptance. We originally anticipated submitting the study for Stage 2 review one year after in-principle acceptance. However, this timeline required adjustments due to several factors. First, the stimuli underwent additional modifications, which extended the preparation phase. Additionally, setting up the study for multiple eye-trackers and their respective software further contributed to delays. Furthermore, varying COVID-19 restrictions such as closures and reopening schedules across different labs introduced additional challenges and delayed the start of data collection. The most significant source of delay, however, was the data pre-processing phase, which took longer than expected. These combined factors necessitated a longer timeline before submission for Stage 2 review.

Participation Criterion

The participating labs were recruited from the MB2 consortium. In July 2020, we asked via the MB2 listserv which labs planned to contribute how many participants for the respective age group (toddlers and/or adults). Each lab made a commitment to collecting data from at least 16 participants (toddlers or adults), but we did not exclude any contributed data on the basis of

the total sample size contributed by that lab. Labs were allowed to test using either in-lab eye-tracking or online methods.

Ethics

All labs were responsible for obtaining ethics approval from their appropriate institutional review board. The labs contributed de-identified data for central data analysis (i.e., eye-tracking raw data/coded gaze behavior, demographic information). Video recordings of the participants were stored at each lab according to the approved local data handling protocol. If allowed by the local institutional review board, video recordings were made available to other researchers via the video library Databrary (<https://nyu.databrary.org/>).

Participants

In a preliminary expression of interest, 26 labs signed up to contribute a minimal sample size of 16 toddlers and/or adults. Based on this information, we expected to recruit a total sample of 520 toddlers (ages 18-27 months) and 408 adults (ages 18-55 years). To avoid an unbalanced age distribution in the toddlers' sample, labs signed up for testing at least one of two age bins (bin 1: 18-22 months, bin 2: 23-27 months), and were asked to ensure approximately equal distribution of participants' age in their collected sample if possible. They were asked to try to ensure that the mean age of their sample lies in the middle of the range of the chosen bin and that participant ages were distributed across their whole bin. Both for adults and toddlers, basic demographic data was collected on a voluntary basis with a brief questionnaire (see Supplemental Material for details). The requested demographic information that was not used in the registered confirmatory and/or exploratory analyses of this study has been collected for further potential follow-up analyses in spin-off projects within the MB framework.

After completing the task, adult participants were asked to fill a funneled debriefing questionnaire. This questionnaire asked what the participant thought the purpose of the

experiment was, whether the participant had any particular goal or strategy while watching the videos, and whether the participant consciously tracked the chaser's epistemic state. Additionally, we collected details regarding each testing session (see Supplemental Material S3).

Our final dataset consisted of 1224 participants, with an overall exclusion rate of 24.16% (toddlers: 35.60%, adults: 12.67%). Table 1 and Table 2 show the distribution of included participants across labs, eye-tracking methods, and ages. A final sample of 521 toddlers (49.14% female) that were tested in 37 labs (mean lab sample size = 14.08, SD = 5.56, range: 2 - 32) was analyzed. The average age of toddlers in the final sample was 22.49 months (SD = 2.53, range: 18 - 27.01). The final sample size of included adults was 703 (68.85% female), tested in 34 labs (mean lab sample size = 20.68, SD = 12.14, range: 8 - 65). Their mean age was 24.61 years (SD = 7.36, range: 18 - 55).

Table 1

Lab and Participant information for the adult age cohort

Lab	<i>n</i> collected	<i>n</i> included	Sex (<i>n</i> Female)	Mean Age (years)	Method
CogConcordia	21	16	11	22.12	In-lab
CorbitLab	16	15	14	19.87	In-lab
DevlabAU	20	20	15	25.15	In-lab
MEyeLab	53	53	39	24.47	In-lab
MiniDundee	15	13	10	30.23	In-lab
PKUSu	39	32	19	22.66	In-lab
SkidLSDLab	11	8	3	21.62	In-lab
ToMcdlSalzburg	33	31	22	27.23	In-lab
UIUCinfantlab	36	32	25	19.06	In-lab
WSUMARCS	18	13	8	29.85	In-lab
affcogUTSC	23	8	5	20.88	Web-based
babyLeidenEdu	20	16	12	23.31	In-lab
babylabAmsterdam	17	16	13	24.00	In-lab
babylabBrookes	67	65	49	21.78	In-lab
babylabINCC	18	18	12	31.00	In-lab
babylabMPIB	16	16	11	27.44	In-lab
babylabNijmegen	19	15	13	22.13	In-lab
babylabTrento	16	16	9	21.69	In-lab

Lab	<i>n</i> collected	<i>n</i> included	Sex (<i>n</i> Female)	Mean Age (years)	Method
babylabUmassb	33	11	10	19.00	In-lab
babyuniHeidelberg	16	16	14	22.06	In-lab
beinghumanWroclaw	19	16	9	32.75	Web-based
careylabHarvard	18	15	12	19.80	In-lab
ccIUNIRI	32	32	17	30.53	In-lab
childdevlabAshoka	16	16	8	30.88	In-lab
collabUIOWA	16	16	10	19.19	In-lab
gaugGöttingen	30	28	18	31.71	In-lab
jmuCDL	32	32	22	18.81	In-lab
kidsdevUniofNewcastle	15	14	7	33.57	In-lab
labUNAM	20	11	8	22.45	In-lab
ImuMunich	31	30	23	22.53	In-lab
mecdmpihcbs	19	19	10	27.79	In-lab
socialcogUiami	16	15	9	19.27	In-lab
sociocognitivelab	17	17	11	32.12	In-lab
tauccd	15	12	6	24.50	In-lab
Total	803	703	484	24.75	

Table 2*Lab and Participant information for the toddler age cohort*

Lab	<i>n</i> collected	<i>n</i> included	Sex (<i>n</i> Female)	Mean Age (months)	Method
CogConcordia	21	8	4	22.92	Web-based
CorbitLab	11	10	5	22.77	In-lab
DevlabAU	18	17	8	19.00	In-lab
PKUSu	50	32	13	20.84	In-lab
SkidLSDLab	8	2	0	20.11	In-lab
ToMcdlSalzburg	17	12	6	22.20	In-lab
UIUCinfantlab	18	15	9	21.96	In-lab
babyLeidenEdu	18	12	8	22.59	In-lab
babylabAmsterdam	28	12	6	23.19	In-lab
babylabBrookes	17	12	7	22.15	In-lab
babylabChicago	17	13	4	20.10	In-lab
babylabINCC	16	9	6	23.40	In-lab
babylabNijmegen	19	10	3	23.52	In-lab
babylabOxford	25	19	8	23.42	In-lab
babylabPrinceton	17	11	7	22.15	In-lab
babylabTrento	18	17	10	22.72	In-lab
babylabUmassb	7	6	2	20.35	In-lab
babylingOslo	17	14	7	21.99	In-lab
babyuniHeidelberg	16	12	4	22.69	In-lab
beinghumanWroclaw	24	14	7	23.77	Web-based
careylabHarvard	17	12	5	21.99	In-lab
cecBYU	16	14	4	22.39	In-lab
childdevlabAshoka	16	10	6	22.44	In-lab

Lab	<i>n</i> collected	<i>n</i> included	Sex (<i>n</i> Female)	Mean Age (months)	Method
gaugGöttingen	28	15	9	23.06	In-lab
gertlabLancaster	21	17	8	23.03	In-lab
infantcogUBC	26	19	8	24.39	In-lab
irlConcordia	19	12	5	22.47	In-lab
kidsdevUniofNewcastle	16	14	9	22.36	In-lab
kokuHamburg	19	14	7	25.99	In-lab
labUNAM	18	12	7	22.68	In-lab
lmuMunich	48	24	16	22.68	In-lab
mecdmpihcbs	25	12	8	23.58	In-lab
mpievaCCP	22	18	10	23.33	In-lab
saxelab	31	15	2	23.13	Web-based
socallabUCSD	47	15	4	22.09	Web-based
tauced	15	12	8	22.99	In-lab
unicph	43	29	16	21.50	In-lab
Total	809	521	256	22.48	

Apparatus and Procedure

Eye-tracking Methods

We expected that participating labs would use one of three types of eye-tracker brands to track the participant's gaze patterns: Tobii, EyeLink, or SMI. Thus, apparatus setup varied slightly across individual labs (e.g., different sampling rates and distances at which the participants were seated in front of the monitor). Participating labs reported their eye-tracker specifications and study procedure alongside the collected data. To minimize variation between labs, all labs using the same type of eye-tracker used the same presentation study file specific to that eye-tracker type. The Supplemental Material provides an overview of employed eye-trackers, stimulus presentation softwares, sampling rates, and screen dimensions.

Online Gaze Recording

To allow for the participation of labs that did not have access to an eye-tracker, or were not able to invite participants to their facilities due to current restrictions regarding the COVID-19 pandemic, labs could choose to collect data via online testing. We initially anticipated for labs

to flexibly choose between different methods for coding gaze direction, including manual frame-by-frame coding from video recordings or using various online platforms for virtual data collection (e.g., LookIt, YouTube, Zoom, Labvanced). Additionally, we considered the possibility of webcam eye-tracking using tools such as WebGazer.js (Papoutsaki et al., 2016). However, we later decided that all labs would conduct only supervised testing, leading to the exclusive use of the MB-ManyWebcams WebGazer setup for data collection (Steffan & Zimmer et al., 2023; see also Supplemental Material S2). This ensured a standardized approach across all participating labs.

Testing Procedure

Toddlers were seated either on their caregiver's lap or in a highchair. The distance from the monitor depended on the data collection method. Caregivers were asked to refrain from interacting with their child and closed their eyes during stimulus presentation or wore a set of opaque sunglasses. Adult participants were seated on a chair at the appropriate distance from the monitor. Once the participant was seated, the experimenter initiated the eye-tracker-specific calibration procedure. Additionally, we presented another calibration stimulus before and after the presentation of the task. This allowed for evaluating the accuracy of the calibration procedure across labs (cf., Frank et al., 2012).

General Lab Practices

To ensure standardization of our experimental procedure, materials for testing practices and instructions were prepared and distributed to the participating labs. Each lab was responsible for maintaining these practices and reported all relevant details on testing sessions (for details see the Supplemental Material).

Videos of Participants

As with all MB projects, we strongly encouraged labs to record video data of their own lab procedures and each testing session, provided that this was in line with regulations of the respective institutional ethics review board and the given informed consent. Participating labs that could not contribute participant videos were asked to provide a video walk-through of their experimental set-up and procedure instead. If no institutional ethics review board restrictions occurred, labs were encouraged to share video recordings of the test sessions via Databrary.

Design Analysis

Here we provide a simulation of the predicted findings because a traditional frequentist power analysis was not applicable for our project for two reasons. First, we used Bayesian methods to quantify the strength of our evidence for or against our hypotheses, rather than assessing the probability of rejecting the null hypothesis. In particular, we computed a Bayes factor (BF; a likelihood ratio comparing two competing hypotheses), which allowed us to compare models. Second, because of the many-labs nature of the study, the sample size was not determined by power analysis, but by the amount of data that participating labs were able to contribute within the pre-established timeframe. Even if the effect size was much smaller than we anticipated (e.g., less than Cohen's $d = 0.20$), the results would have been informative as our study was expected to be dramatically larger than any previous study in this area. If, due to unforeseen reasons, the participating labs were not able to collect a minimum number of 300 participants per age group within the proposed time period, we planned to extend the time for data collection until this minimum number was reached. Conversely, if the effect size was large (e.g., more than Cohen's $d = 0.80$), the resulting increased precision of our model would allow us

to test a number of other theoretically and methodologically important hypotheses (see Results section).

Although we did not determine our sample size based on a power analysis, here we provide a simulation-based design analysis to demonstrate the range of BFs we might have expected to see, given a plausible range of effect sizes and parameters. We focus on our key analysis of the test trials (as specified below), namely the difference in AL on the first test trial that participants saw. We describe below the simulation for the child sample, but based on our specifications, we expected that a design analysis for adult data would produce similar results.

We first ran a simulation for the first look analysis. In each iteration of our simulation, we used a set of parameters to simulate an experiment, using a first look (described below) as the key measure. For the key effect size parameter for condition (knowledge vs. ignorance), we sampled a range of effect sizes in logit space spanning from small to large effects (Cohen's $d = 0.20 - 0.80$; log odds from 0.36 - 1.45). For each experiment, the betas for age and the age x condition interaction were sampled uniformly between -0.20 and 0.20. The age of each participant was sampled uniformly between 18 and 27 months and then centered. The intercept was sampled from a normal distribution (1, 0.25), corresponding to an average looking proportion of 0.73. Lab intercepts and the lab slope by condition were set to 0.1, and other lab random effects were set to 0 as we do not expect them to be meaningfully non-zero. These values were chosen based on pilot data (average looking proportion), but also to have a large range of possible outcomes (lab intercept, age and age x condition interaction). We are confident that the results would be robust to different choices. We then used these simulated data to simulate an experiment with 22 labs and 440 toddlers and computed the resulting BFs, as specified in the analysis plan below. We

adopted all of the priors specified in the results section below³. We ran 349 simulations and, in 72% of them, the BF showed strong evidence in favor of the full model ($BF > 10$); in 6% the BF showed substantial evidence ($10 > BF > 3$); it was inconclusive 14% of the time ($1/10 < BF < 3$), and in 8% of cases the null model was substantially favored (see Supplemental Material Figure S5). The BF was not $< 1/10$ in any of the simulations. Thus, under the parameters chosen here for our simulations, it was likely that the planned experiment was of sufficient size to detect the expected effect.

We also ran a design analysis for the proportional looking analysis. We used the same experimental parameters (number of labs, participants, ages, etc.). For generating simulated data, we drew the condition effect from a uniform distribution between .05 and .20 (in proportion space). The age and age x condition interaction effects were drawn from uniform distributions between -.05 and .05. Sigma, the overall noise in the experiment, was drawn from a uniform distribution between .05 and .1. The intercept was drawn from a normal distribution with mean .65 and a standard deviation of .05. The by-lab standard deviation for the intercept and condition slope was set to .01. Priors were as described in the main text. We ran 119 simulations, and in all 119 we obtained a BF greater than 10, suggesting that, under our assumptions, the study was well-powered.

³ After the design analysis, additional labs expressed their interest in contributing data, which is why the anticipated sample sizes and the numbers this design analysis is based on differ. Given the uncertainty in determining the final sample size in this project, we kept the design analysis as is to have a more conservative estimate of the study's power.

Data Preprocessing

Eye-tracking

Raw gaze position data (x- and y-coordinates) was extracted in the time window starting from the first frame at which the chaser entered the tunnel until the last frame before it exited the tunnel in the familiarization and test trials. For data collected from labs using a binocular eye-tracker, gaze positions of the left and the right eye were averaged.

We used the peekds R package (<https://github.com/peekbank/peekbankr>) to convert eye-tracking data from disparate trackers into a common format. Because not all eye-trackers recorded data with the same frequency or regularity, we resampled all data to be at a common rate of 40 Hz (samples per second).

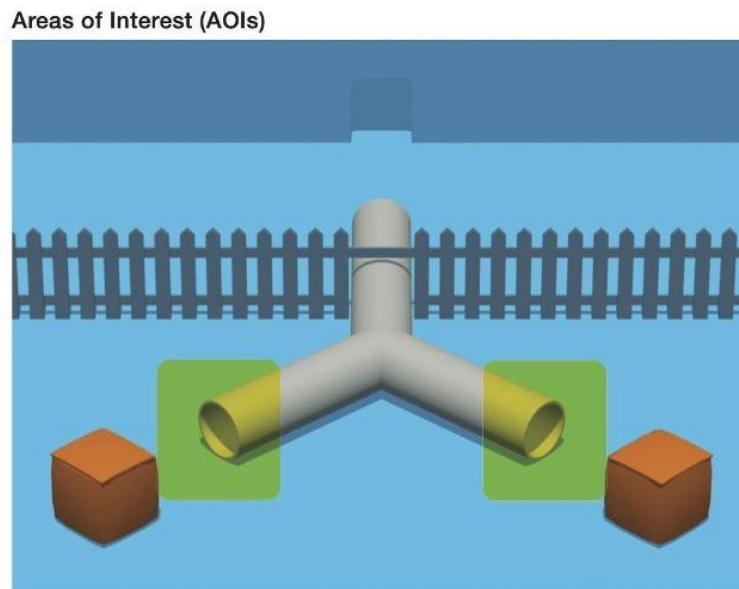
We excluded individual trials if more than 50% of the gaze data was missing (defined as off-screen or unavailable point of gaze during the whole trial, not just the anticipatory period). Applying this criterion would have caused us to exclude 4% of the trials in our pilot data, which inspection of our pilot data suggested was an appropriate trade-off between not excluding too much usable data and not analyzing trials which were uninformative.

For each monitor size, we determined the specific AOIs and computed whether the specific x- and y-position for each participant, trial, and time point fell within their screen resolution-specific AOIs. Our goal was to determine whether participants were anticipating the emergence of the chaser from one of the two tunnel exits. Thus, we defined AOIs on the stimulus by creating a rectangular region around the tunnel exit that was D units from the top, bottom, left, and right of the boundary of the tunnel exit, where D was the diameter of the tunnel exits. We then expanded the sides of the AOI rectangles by 25% in all directions to account for tracker calibration error. Our rationale was that, if we made the AOI too small, we might fail to capture anticipations by participants with poor calibrations. In contrast, if we made the regions too large,

we might capture some fixations by participants looking at the box where the chasee actually was. On the other hand, these chasee looks would not be expected to vary between conditions and so would only affect our baseline level of looking. Thus, the chosen AOIs aimed at maximizing our ability to capture between-condition differences. For an illustration of the tunnel exit AOIs see Figure 3. We were not analyzing looks to the boxes, since they can less unambiguously be interpreted as epistemic state-based action predictions and because we observed few anticipatory looks to the boxes in the pilot studies. For more detailed information about the AOI definition process see the description of the pilot study results in the Supplemental Material.

Figure 3

Illustration of Areas of Interest (AOIs) for gaze data analysis during the anticipatory period



Note. The light green rectangles show the dimensions of the AOIs used for the analysis of AL during the test period.

Manual Coding

For data gathered without an eye-tracker (e.g., videos of participants gathered from online administration), precise estimation of looks to specific AOIs were not possible. Instead, videos were coded for whether participants were looking to the left or the right side of the screen (or “other/off screen”). In our main analysis, during the critical anticipatory window, we treated these looks identically to looks to the corresponding AOI. See exploratory analyses for analysis of data collected online.

Temporal Region of Interest

For familiarization trials, we defined the start of the anticipatory period (total length = 4000 ms) as starting 120 ms after the first frame after which the chaser had completely entered the tunnel and lasting until 120 ms after the first frame at which the chaser was visible again (we chose 120 ms as a conservative value for cutting off reactive saccades; cf., Yang et al., 2002). For test trials, we defined the start of the anticipatory period in the same way, with a total duration of 4000 ms.

Dependent Variables

We define two primary dependent variables:

1. First look. First saccades were determined as the first change in gaze occurring within the anticipatory time window that was directed towards one of the AOIs. The first look was then the binary variable denoting the target of this first saccade (i.e., either the correct or incorrect AOI) and was defined as the first AOI where participants fixated at for at least 150 ms, as in Rayner et al. (2009). The rationale for this definition was that, if participants were looking at a location within the tunnel exit AOIs before the anticipation period, they might have been looking there for other reasons than action prediction. We therefore

counted only looks that started within the anticipation period because they more unambiguously reflected action predictions. This further prevented us from running into a situation where we would have included a lot of fixations on regions other than the tunnel exit AOIs because participants were looking somewhere else before the anticipation period began.

2. Proportion DLS (also referred to as total relative looking time; Senju et al., 2009). We computed the proportion looking (p) to the correct AOI during the full 4000 ms anticipatory window (correct looking time / (correct looking time + incorrect looking time)), excluding looks outside of either AOI.

Results

Confirmatory Analyses

Approach

We adopted a Bayesian analysis strategy to maximize our ability to make inferences about the presence or absence of a condition effect (i.e., our key effect of interest). In particular, we fitted Bayesian mixed effects regressions using the package *brms* in R (Bürkner, 2017). This framework allowed us to estimate key effects of interest while controlling for variability across grouping units (in our case, labs).

To facilitate interpretation of individual coefficients, we report means and credible intervals. For key inferences in our confirmatory analysis, we used the bridge sampling approach (Gronau et al., 2017) to compute BFs comparing different models. As the ratio of the likelihood of the observed data under two different models, BFs allowed us to quantify the evidence that our

data provide with respect to key comparisons. For example, by comparing models with and without condition effects, we can quantify the strength of the evidence for or against such effects.

Bayesian model comparisons require the specification of proper priors on the coefficients of individual models. Here, for our first look analysis, we used a set of weakly informative priors that capture the expectation that the effects that we observe (of condition and, in some cases, trial order) are modest. For coefficients, we chose a normal distribution with mean of 0 and *SD* of 2. Based on our pilot testing and the results of MB1, we assumed that lab and participant-level variation would be relatively small, and so for the standard deviation of random effects (i.e., variation in effects across labs and, in the case of the familiarization trials, participants) we set a Normal prior with mean of 0 and *SD* of 0.1. We set an LKJ(2) prior on the correlation matrix in the random effect structure, a prior that is commonly used in Bayesian analyses of this type (Bürkner, 2017). Because the BF is sensitive to the choice of prior, we also ran a secondary analysis with a less informative prior: fixed effect coefficients chosen from a normal distribution with mean 0 and *SD* of 3, and random effect standard deviations drawn from a normal prior with a mean of 0 and *SD* of 0.5 (see Supplemental Material S4). With respect to the specification of random effects, we followed the approach advocated by Barr (2013), that is, specifying the maximal random effect structure justified by our design. Since we were interested in lab-level variation, we fitted random effect coefficients for fixed effects of interest within labs (e.g., condition within lab). Further, where there were participant-level repeated measure data (e.g., familiarization trials), we fitted random effects of participants.

For the proportion DLS analysis, we used a uniform prior on the intercept between -0.5 and 0.5 (corresponding to proportional looking scores between 0 and 1: the full possible range). For the priors on the fixed effect coefficients, we used a normal prior with a mean of 0 and an *SD* of 0.1. Because these regressions are in proportion space, 0.10 corresponds to a change in

proportion of 10%. For the random effect priors, we used a normal distribution with mean 0 and standard deviation .05. The LKJ prior was specified as above.

All preregistered confirmatory models converged, with no divergent transitions, all Rhat values < 1.1, and an effective sample size greater than 20% of the total sample size.

Familiarization Trials

Figure 4 shows first look and proportion DLS (non-logit transformed) for toddlers and adults plotted across familiarization trials and test trials. Our first set of analyses examined data from the four familiarization trials and asked whether participants anticipated the chaser's reappearance at one of the tunnel exits. In our first analysis, we were interested in whether participants engaged in AL during the familiarization trials. To quantify the level of familiarization, we fitted Bayesian mixed effect models predicting target looks based on trial number (1-4) with random effects for lab and participants and random slopes for trial number for each.

In R formula notation (which we adopt here because of its relative concision compared with standard mathematical notation), our base model was as follows:

$$measure \sim 1 + trial_{number} + (trial_{number}|lab) + (trial_{number}|participant).$$

We fitted a total of four instances of this model, one for each age group (toddlers vs. adults) and dependent measure (first look vs. proportion DLS). First look models were fitted using Bernoulli family with a logit link function. The proportion DLS models were Gaussian and the dependent variable was centered by subtracting 0.5, such that 0 corresponded to chance-level performance.

Our key question of interest was whether overall anticipation was higher than chance levels on the familiarization trial immediately before the test trials, in service of evaluating the evidence that participants were attentive and making predictive looks immediately prior to test.

To evaluate this question across the four models, we coded trial number so that the last trial before the test trials (trial 4) was set to the intercept, allowing the model intercept to encode an estimate of the proportion of correct anticipation immediately before test. We then fitted a simpler model for comparison

$$measure \sim 0 + trial_number + (trial_number | lab) + (trial_number | participant),$$

which included no intercept term. We then computed the BF comparing this model to the full model. This BF quantified the evidence for an anticipation effect for each group and measure.

First Look.

Toddlers. Investigating first looks to the target location for toddlers, we used a Bayesian mixed effects model to predict whether toddlers' first look was to the target exit based on trial number (1-4), with random effects for lab and participants and random slopes for trial number for each. The Bayes factor comparing the full model to the simpler model was estimated to be $BF > 1000$, favoring the full model over the null model. This confirmed that toddlers showed above-chance looking to the target location during the anticipatory window of the last familiarization trial. The model also provided support for an effect of trial number on proportion of first looks, with the negative coefficient indicating a decrease in first looks to the target location across the familiarization trials.

Adults. Comparing the Bayesian mixed effects model of adults predicting proportion of first looks based on trial number (1-4), with random effects for lab and participants and random slopes for trial number for each with the simpler model without an intercept, we computed a Bayes factor of $BF > 1000$, strongly favoring the full model over the null model. Again, this suggested that also adults' anticipation was higher than chance levels on the last familiarization trial. In addition, there was support for an effect of trial number on proportion of first looks, with

the positive coefficient indicating an increase in proportion of first target looks across the familiarization trials.

Proportion DLS.

Toddlers. We used a Bayesian mixed effects model to predict proportion DLS based on trial number (1-4) for toddlers, with random effects for lab and participants and random slopes for trial number for each. The Bayes factor comparing this model to the simpler null model without the intercept was estimated to be $BF > 1000$, strongly favoring the full model over the null model. See also Table 3 for regression coefficients for the full model. These results suggest a significant effect of trial number on proportion DLS, with the negative coefficient indicating a decrease in proportion DLS across the familiarization trials.

Adults. Next, we used a Bayesian mixed effects model to predict proportion DLS based on trial number (1-4) for adults, again with random effects for lab and participants and random slopes for trial number for each. The Bayes factor for the full model against the null model was $BF > 1000$, suggesting strong evidence for the full model. These results suggest a significant effect of trial number on proportion DLS, with the positive coefficient indicating an increase in target looks across the familiarization trials.

Test Trials

We focused our confirmatory analysis on the first test trial (see Exploratory Analysis section for an analysis of both trials). Our primary question of interest was whether AL differs between conditions (knowledge vs. ignorance, coded as $-.5/.5$) and by age (in months, centered). For child participants, we fitted models with the specification:

$$\begin{aligned} \text{measure} \sim & 1 + \text{condition} + \text{age} + \text{condition:age} \\ & + (1 + \text{condition} + \text{age} + \text{condition:age} | \text{lab}). \end{aligned}$$

For adult participants, we fitted models with the specification:

$$measure \sim 1 + condition + (1 + condition|lab).$$

Again, we fitted models with a logistic link for first look analyses and with a standard linear link for proportion DLS.

In each case, our key BF was a comparison of this model with a simpler “null” model that did not include the fixed effect of condition but still included other terms. We took a $BF > 3$ in favor of a particular model as substantial evidence and a $BF > 10$ in favor of strong evidence. A $BF < 1/3$ was taken as substantial evidence in favor of the simpler model, and a $BF < 1/10$ as strong evidence in favor of the simpler model.

For the model of data from toddlers, we additionally were interested in whether the model showed changes in AL with age. We assessed evidence for this by computing BFs related to the comparison with a model that did not include an interaction between age and condition as fixed effects

$$measure \sim 1 + condition + age + (1 + condition + age + condition:age|lab).$$

These BFs captured the evidence for age-related changes in the difference in action anticipation between the two conditions.

It is important to note that in the case of a null effect, there are two main explanations: (1) toddlers and adults in our study do not distinguish between knowledgeable and ignorant agents when predicting their actions. (2) The method used is not appropriate to reveal knowledge/ignorance understanding. By using Bayesian analyses, we are able to better evaluate the first of these two possibilities: The BF provides a measure of our statistical confidence in the null hypothesis, i.e., no difference between experimental conditions, given the data in ways that standard null hypothesis significance testing does not. In other words, instead of merely concluding that we did not find a difference between conditions, we would be able to find no/anecdotal/moderate/strong/very strong/extreme evidence for the null hypothesis that our

participants did not distinguish between knowledgeable and ignorant agents when predicting their actions (Schönbrodt & Wagenmakers, 2018). We therefore consider this analysis an important addition to our overall analysis strategy. Yet, even our Bayesian analyses are not able to rule out the second possibility that participants may well show such knowledge/ignorance differentiation with different methods, or that this ability may not be measurable with any methods available at the current time. Addressing this alternative explanation warrants follow up experiments.

First look.

Toddlers. Investigating first looks for toddlers, we again used a Bayesian mixed effects model to predict target looks based on condition, with random effects for lab. The Bayes factor comparing the full model to the simpler model was estimated to be $BF = 2.4$, providing only anecdotal evidence in favor of the full model over the null model.

Again, we examined whether age influenced the difference in action anticipation between knowledge and ignorance trials. To do this, we compared the full model, which included an interaction between age and condition, with a simpler model without this interaction. The computed Bayes factor, $BF < 0.01$, extremely supports the simpler model, suggesting that the interaction term does not substantially improve the model's fit. This implies that age does not appear to significantly affect the difference in action anticipation between the two trial types.

Adults. We compared a Bayesian mixed-effects model predicting the proportion of first looks based on condition, including random effects for lab to a simpler model without the main effect of condition. The analysis yielded a Bayes factor of $BF > 1000$, providing strong evidence in favor of the full model over the null model. Results indicated that first looks to the target were significantly more frequent in the knowledge condition compared to the ignorance condition.

Proportion DLS. Figure 5 depicts mean proportion of looking to target and distractor during the anticipation window. The timecourse plot displays mean proportion of looking to

target and distractor for both conditions, knowledge and ignorance, both age cohorts, toddlers and adults, as well as both test trials, first and second.

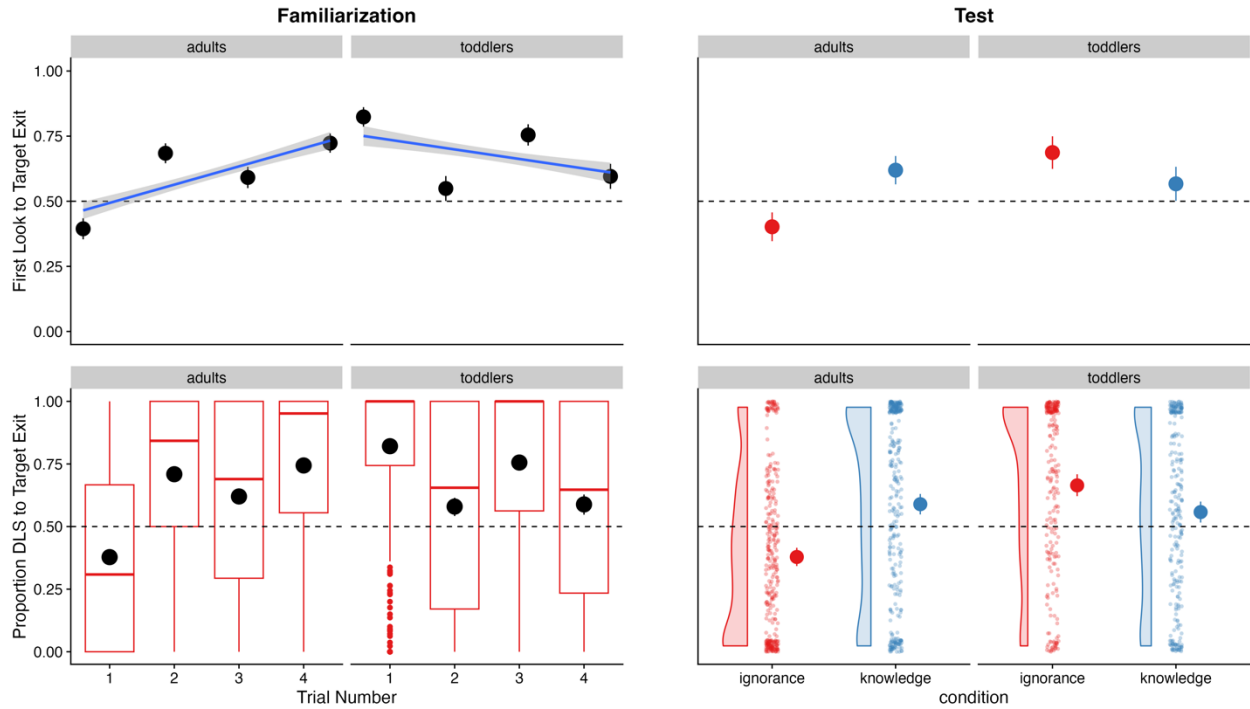
Toddlers. As a first model, we used a Bayesian mixed effects models to predict toddlers' proportion DLS based on condition, age, and the interaction of condition and age, while accounting for variability across labs. The Bayes factor comparing this model to the simpler null model without the interaction of condition was estimated to be $BF = 21.2$, favoring the full model over the null model. Table 4 shows the statistics for regression coefficients of the full model. These results suggest a significant effect of condition on proportion DLS, with the positive coefficient indicating higher proportion DLS for ignorance trials compared to knowledge trials. An exploratory test of above-chance looking in the knowledge condition, however, revealed that toddlers looked significantly above chance to the target, $t(255) = 2.64, p = 0.009$.

Additionally, we assessed whether toddlers' AL changed with age. Comparing our full model, which included an interaction between age and condition, with a simpler model without this interaction, yielded a Bayes factor, $BF = 0.4$, providing only anecdotal evidence supporting the simpler model. This result suggests that the interaction between age and condition might not be a necessary predictor, as it does not provide substantial additional explanatory power. Hence, our results do not provide sufficient evidence to determine whether age-related changes in AL are consistent across conditions or differ between them.

Adults. Next, we used a Bayesian mixed effects model to predict proportion DLS based on condition for adults, again with random effects for lab. The Bayes factor comparing this model to the simpler null model without the main effect of condition was estimated to be $BF > 1000$, strongly favoring the full model over the null model. These results suggest a significant main effect of condition on proportion DLS, with the negative coefficient indicating a higher number of target looks for knowledge than for ignorance trials.

Figure 4

First look and proportion DLS for toddlers and adults during familiarization and test phase



Note. For familiarization, first look and proportion DLS are shown for each of the four trials during the anticipatory window. For the test phase, first look and proportion DLS are shown for the anticipatory window of the first trial only. Error bars and error bands represent 95% CIs.

Table 3

Results of the Bayesian mixed effects models for the familiarization phase

model	term	β	SE	lower CrI95%	upper CrI95%	$\hat{R}$
first look toddlers	Intercept	0.44	0.09	0.27	0.61	1.00
	Trial Number	-0.22	0.05	-0.32	-0.12	1.00
first look adults	Intercept	1.03	0.09	0.86	1.20	1.00
	Trial Number	0.38	0.04	0.30	0.47	1.00
proportion DLS toddlers	Intercept	0.12	0.02	0.09	0.15	1.00
	Trial Number	-0.05	0.01	-0.06	-0.03	1.00
proportion DLS adults	Intercept	0.26	0.02	0.23	0.29	1.00
	Trial Number	0.10	0.01	0.09	0.11	1.00

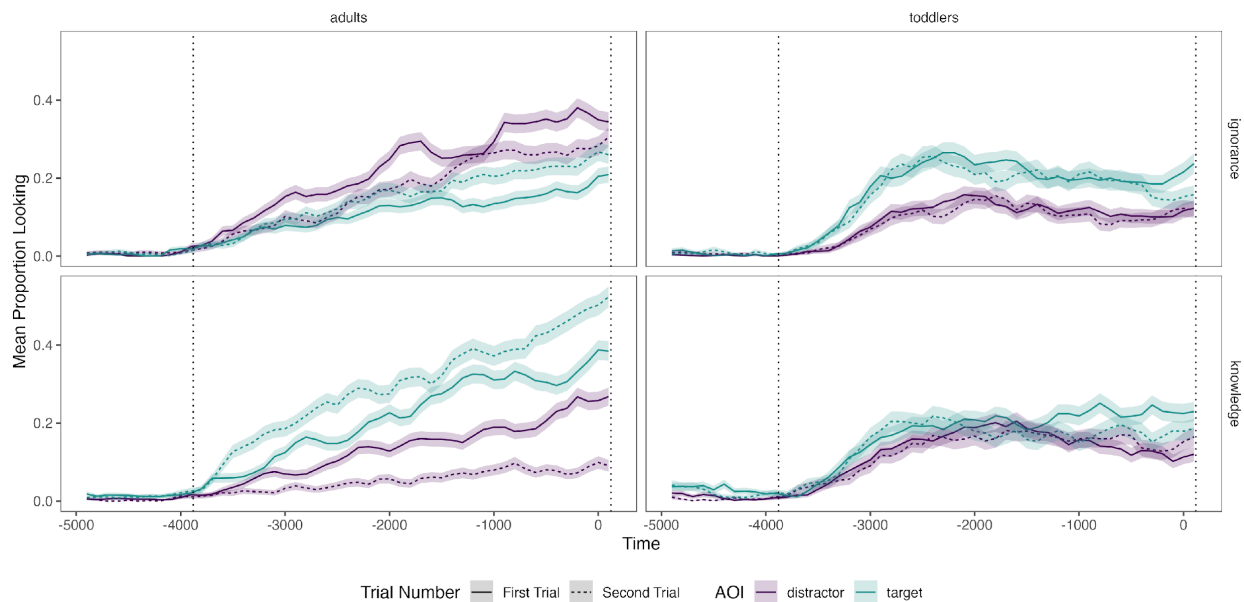
Table 4

Results of the Bayesian mixed effects models for the test phase

model	term	β	SE	lower CrI95%	upper CrI95%	$\hat{R}$
first look toddlers	Intercept	0.53	0.11	0.32	0.74	1.00
	Condition	0.53	0.21	0.13	0.93	1.00
	Age	0.06	0.05	-0.03	0.15	1.00
	Condition:Age	-0.13	0.09	-0.30	0.04	1.00
first look adults	Intercept	0.05	0.09	-0.12	0.22	1.00
	Condition	-0.89	0.17	-1.22	-0.56	1.00
proportion DLS toddlers	Intercept	0.61	0.02	0.58	0.65	1.00
	Condition	0.10	0.03	0.03	0.17	1.00
	Age	0.01	0.01	-0.01	0.02	1.00
	Condition:Age	-0.01	0.02	-0.04	0.02	1.00
proportion DLS adults	Intercept	0.48	0.02	0.45	0.51	1.00
	Condition	-0.20	0.03	-0.26	-0.15	1.00

Figure 5

Timecourse of mean proportion looking to target and distractor for the first and second test trial for toddlers and adults in the ignorance and knowledge condition



Note. The dotted vertical lines represent the onset and offset of the 4s anticipatory period. Time is

centered such that 0 represents the moment the bear re-remerges from the tunnel. Error bands represent ± 1 SEs.

Exploratory Analyses

Spill-over

We analyzed within-participants data from the second test trial that participants saw, using exploratory models to assess whether (1) findings are consistent when both trials are included (overall condition effect), (2) whether effects are magnified or diminished on the second trial (order main effect), and (3) whether there is evidence of “spillover”-dependency in anticipation on the second trial depending on what the first trial is (condition x order interaction effect; see Figure 5 and Figure 6).

For toddlers, we analyzed condition-effects of within-participants data for both test trials by fitting a Bayesian mixed-effects model with the dependent variable of proportion DLS and main effects of condition and age and their interaction. Comparing this full model to a null model that did not include the fixed effect of condition, we obtained a Bayes Factor of $BF = 147.1$, providing strong evidence in favor of the full model, suggesting no within-participants condition effect.

For adults, we also fitted a Bayesian mixed-effects model to predict their proportion DLS for both test trials with the main effect of condition and random effects for participant and lab. Again, the data provided very strong evidence for the inclusion of the main effect of condition with a Bayes Factor of $BF > 1000$. The effect of condition was negative and credible, suggesting that proportion DLS was significantly lower in the ignorance condition compared to the knowledge condition.

In order to investigate whether there was an interaction of condition and test trial number, we fitted Bayesian mixed-effects models to predict proportion DLS with fixed effects for condition, test trial number, and their interaction, along with random intercepts and slopes for these variables across labs, for toddlers and adults separately. For toddlers, the Bayes factor, $BF = 0.4$, provided anecdotal evidence for the simpler null model without the interaction term, indicating that the interaction between condition and test trial number does not add substantial explanatory power to the model. These results suggest that neither condition nor its interaction with test trial number significantly impacted proportion DLS in this sample.

For adults, the Bayes Factor, $BF = 19.7$, provided strong evidence for including the interaction of condition and test trial number as a fixed effect. These results indicate that while proportion DLS increased over trials, this effect was moderated by condition, with the ignorance condition showing a slower rate of increase compared to the knowledge condition.

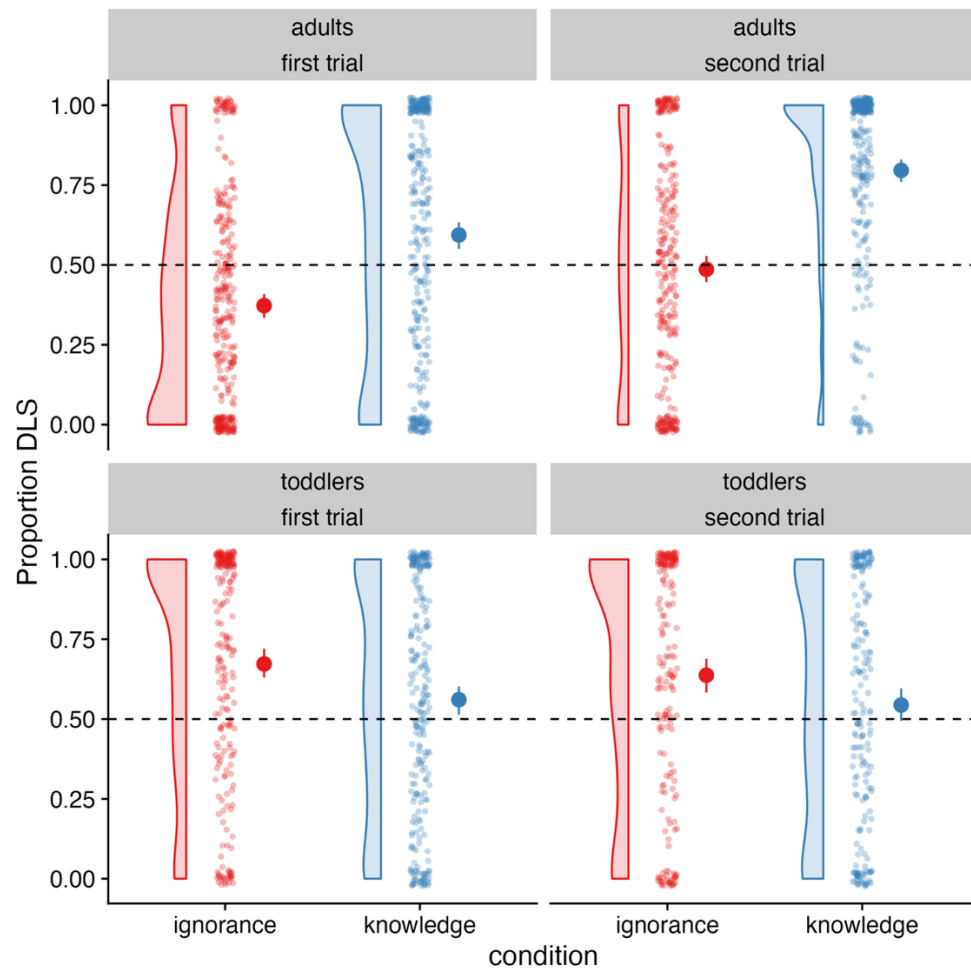
To examine whether anticipatory looking during the second test trial was influenced by condition and anticipatory looking during the first test trial, we fitted a Bayesian mixed-effects model for each age cohort separately. This model included fixed effects for condition, proportion of target looking during the first test trial, and their interaction. Random intercepts and slopes for these predictors were modeled at the lab level.

For toddlers, the Bayes factor, $BF = 1.1$, suggests anecdotal evidence in favor of including these predictors compared to the null model. In addition, we compared the full model to a model excluding the main effect of condition. The Bayes factor was $BF = 2.3$, again providing only anecdotal evidence in favor of the full model that included condition. Taken together, these results suggest that condition and its interaction with first trial anticipatory looking were not strong predictors of second trial anticipatory looking behavior in toddlers.

For adults, the Bayes factor comparing the full model including condition, first trial anticipatory looking, and their interaction to the null model with only the intercept was $BF > 1000$ providing extreme evidence for the full model. Comparing the full model with a model that excluded condition, the Bayes factor, $BF > 1000$, provided again extreme evidence in favor of the full model. Hence, the extremely large Bayes factors underscore the importance of considering these predictors in explaining second test trial AL behavior.

Figure 6

Proportional DLS for the first and second test trial for toddlers and adults in the ignorance and knowledge condition



Note. Error bars represent 95% CIs.

Relation between familiarization and test

We explored whether condition differences varied for participants who showed higher rates of anticipation during the four familiarization trials. To investigate whether participants who showed anticipatory looking during the familiarization phase also exhibit anticipatory looking during the test phase, we explored three different measures. First, we assessed anticipatory looking in participants who successfully anticipated during the final familiarization trial, defined

as those whose first fixation was on the target. Second, we examined anticipatory looking in participants who consistently demonstrated anticipatory behavior across the last two familiarization trials before the test trials, operationalized as having a proportion DLS greater than 0.5 in each trial. Finally, we computed correlations to explore whether performance in the familiarization phase was related to performance in the test trials.

Relation between anticipatory looking during the first test trial and first look during the final familiarization trial. We fitted a main Bayesian hierarchical model testing the fixed effects of condition (ignorance vs. knowledge), first look during the final familiarization trial (target vs. distractor), and their interaction on first-trial proportion DLS during the anticipatory window for toddlers and adults separately. Random intercepts and slopes for all fixed effects and their interaction were included at the lab level, accounting for variability across different experimental settings.

For toddlers, the Bayes factor comparing this model to the simpler null model without the interaction of condition and first look during the final familiarization trial indicated that there was anecdotal evidence in favor of the simpler null model over the full model, $BF = 0.7$. The effect of condition was positive, but its confidence interval narrowly included zero, suggesting weak evidence for a condition effect (see Table 5). The effect of performance during the final familiarization trial was close to zero, indicating no substantial main effect of prior performance. Similarly, the interaction between condition and performance in the final familiarization trial was small and non-significant. These results suggest that while there was some weak evidence for a main effect of condition on anticipatory looking, neither performance during the final familiarization trial nor its interaction with condition substantially predicted anticipatory looking during the test trial. This result indicates that the relation between anticipatory looking during the first test trial and condition did not depend significantly on prior familiarization performance.

For adults, the Bayes factor comparing this model to the simpler null model without the main effect of condition was estimated to be $BF > 1000$, strongly favoring the full model over the null model. The regression coefficients showed a significant negative effect of condition, indicating that anticipatory looking was lower in ignorance trials compared to knowledge trials. The decisive Bayes factor strongly favors the inclusion of condition and familiarization trial performance in the model, suggesting that these predictors are relevant for understanding anticipatory looking in adults. However, the small and non-significant estimates for the effects of familiarization trial performance and its interaction with condition imply that condition is the primary driver of anticipatory looking differences, with performance in familiarization trials contributing minimally.

Only above 50% looking to target during familiarization trials. To examine the effect of condition and successful anticipatory looking during familiarization (above 50% target looking during the last two familiarization trials before test) on anticipatory looking during the first test trial, we fitted Bayesian mixed-effects models for each age group separately. The models included fixed effects for condition, successful anticipatory looking during familiarization trials 3 and 4, and their interaction. Random intercepts and slopes for these predictors were included at the lab level.

Comparing the full model to the null model of toddlers revealed a Bayes Factor of $BF = 17.9$, providing strong evidence favoring the full model over a null model that excludes these predictors, suggesting that these factors contribute meaningfully to explaining the variance in test trial anticipatory looking. The regression analysis showed a positive main effect of condition, indicating higher anticipatory looking in one condition compared to the other (see Table 5). There was a small positive, but non-significant, effect of successful anticipatory looking during familiarization. The interaction between condition and successful anticipatory looking during

familiarization was also small and non-significant. These results indicate that condition is a meaningful predictor of anticipatory looking during test trials in toddlers, with participants showing different levels of anticipatory looking based on condition. However, the successful anticipatory looking during familiarization trials and its interaction with condition appear to have minimal additional impact. The strong Bayes factor further supports the importance of including these predictors in the model but highlights that condition remains the primary driver of test trial differences.

The estimated Bayes factor in favor of the full model of adults over the null model was $BF > 1000$, indicating that the predictors substantially contribute to explaining test trial anticipatory looking. The regression coefficients revealed a significant main effect of condition, with participants showing lower anticipatory looking in the ignorance condition compared to the knowledge condition. There was a small, positive, and non-significant effect of successful anticipatory looking during familiarization and the interaction between condition and successful anticipatory looking during familiarization was negligible. These results indicate that condition has a substantial and meaningful impact on anticipatory looking during the first test trial in adults, while successful anticipatory looking in familiarization trials and its interaction with condition have limited additional influence. The extremely large Bayes factor highlights the strong explanatory power of including these predictors in the model, although condition remains the primary driver of the observed differences.

Table 5

Results of the Bayesian mixed effects models for the relation between familiarization and test

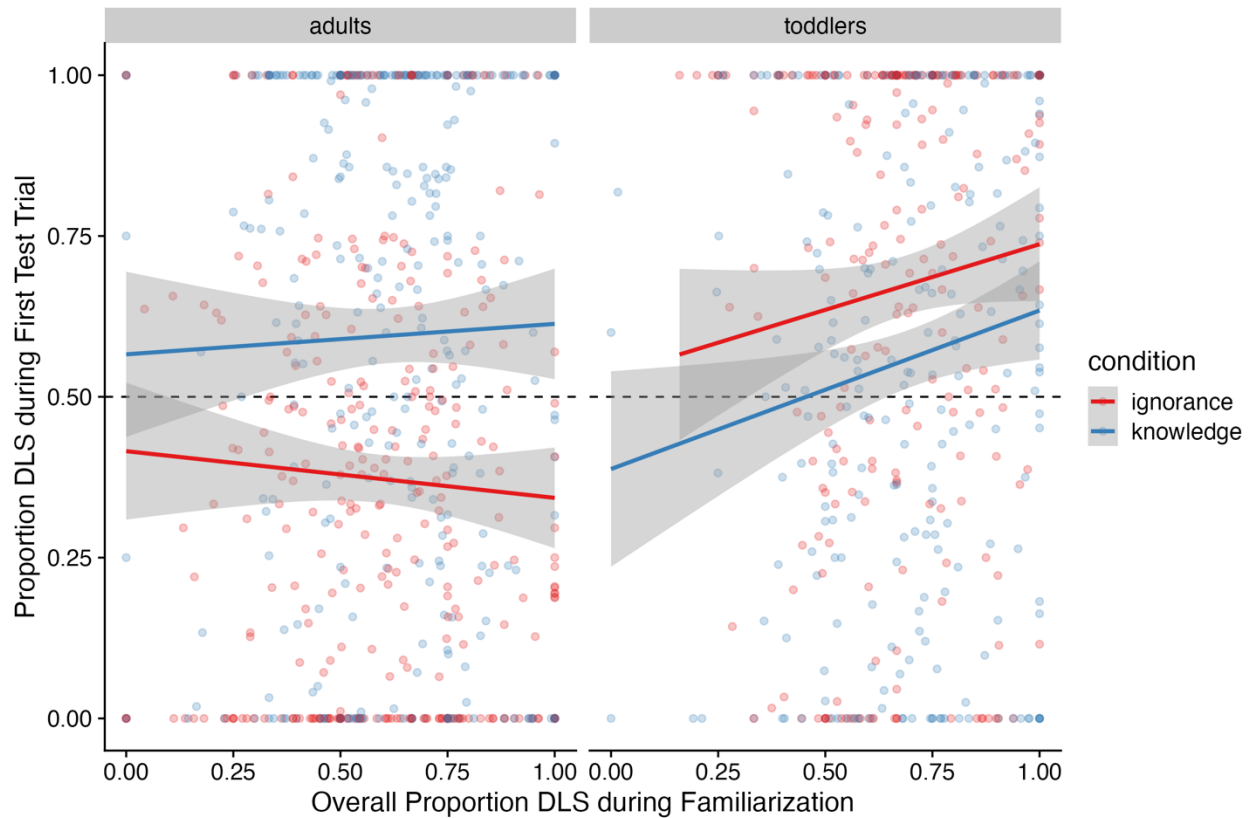
model	term	β	SE	lower CrI95%	upper CrI95%	$\hat{R}$
sufficient familiarization adults	Intercept	0.48	0.02	0.44	0.51	1.00
	Condition	-0.20	0.03	-0.26	-0.14	1.00
	Successful Fam Anticipation	0.04	0.03	-0.02	0.10	1.00
	Condition:Successful Fam Anticipation	0.00	0.05	-0.11	0.10	1.00
sufficient familiarization toddlers	Intercept	0.61	0.02	0.56	0.65	1.00
	Condition	0.10	0.04	0.02	0.17	1.00
	Successful Fam Anticipation	0.03	0.04	-0.04	0.11	1.00
	Condition:Successful Fam Anticipation	0.02	0.06	-0.11	0.14	1.00
correct first look adults	Intercept	0.48	0.02	0.45	0.52	1.00
	Condition	-0.20	0.03	-0.26	-0.13	1.00
	Correct First Look Final Fam	-0.04	0.03	-0.10	0.03	1.00
	Condition:Correct First Look Final Fam	-0.08	0.06	-0.19	0.04	1.00
correct first look toddlers	Intercept	0.62	0.02	0.58	0.65	1.00
	Condition	0.07	0.04	0.00	0.15	1.00
	Correct First Look Final Fam	0.00	0.04	-0.07	0.07	1.00
	Condition:Correct First Look Final Fam	-0.05	0.06	-0.17	0.07	1.00

Correlation between familiarization and test. We also examined the correlation between familiarization and test performance across the two age cohorts and conditions (see Figure 7). While no significant correlations were found for adults in either condition, toddlers in the knowledge condition exhibited a significant positive correlation of anticipatory looking in familiarization and test, $r = 0.15$, $t(254) = 2.35$, $p = 0.02$, suggesting that toddlers, who showed higher proportion DLS overall during familiarization, also showed higher proportion DLS in the first test trial. No such significant correlation was found for toddlers in the ignorance condition, $r = 0.11$, $t(234) = 1.68$, $p = 0.09$. However, there was also no significant interaction between

condition and the correlation in anticipatory looking between familiarization and test for toddlers ($p = .79$).

Figure 7

Relation between anticipatory looking during familiarization and test for both age cohorts and conditions



Note. Error bars represent 95% CIs.

Looking patterns during mouse's change of location

To examine whether participants monitored both the bear and the mouse during the mouse's location change, and whether this influenced AL in the test phase, we defined new time windows of interest (TOIs) corresponding to the mouse's location change in each condition and new AOIs for both the mouse and bear. We hypothesized that participants who attended to both

AOIs would exhibit greater AL compared to those who predominantly tracked the mouse during its location change. Specifically, we analyzed the frequency of gaze shifts between the mouse and bear mouse's location change. An additional exploratory analysis of differential gaze duration directed toward mouse and bear during the mouse's location change is provided in the Supplemental Material S4. Heatmap videos for knowledge and ignorance trials of both age cohorts can be found at https://drive.google.com/drive/folders/1vL-sobdpdGAirqWkr_ZB3eEw6gApvL3.

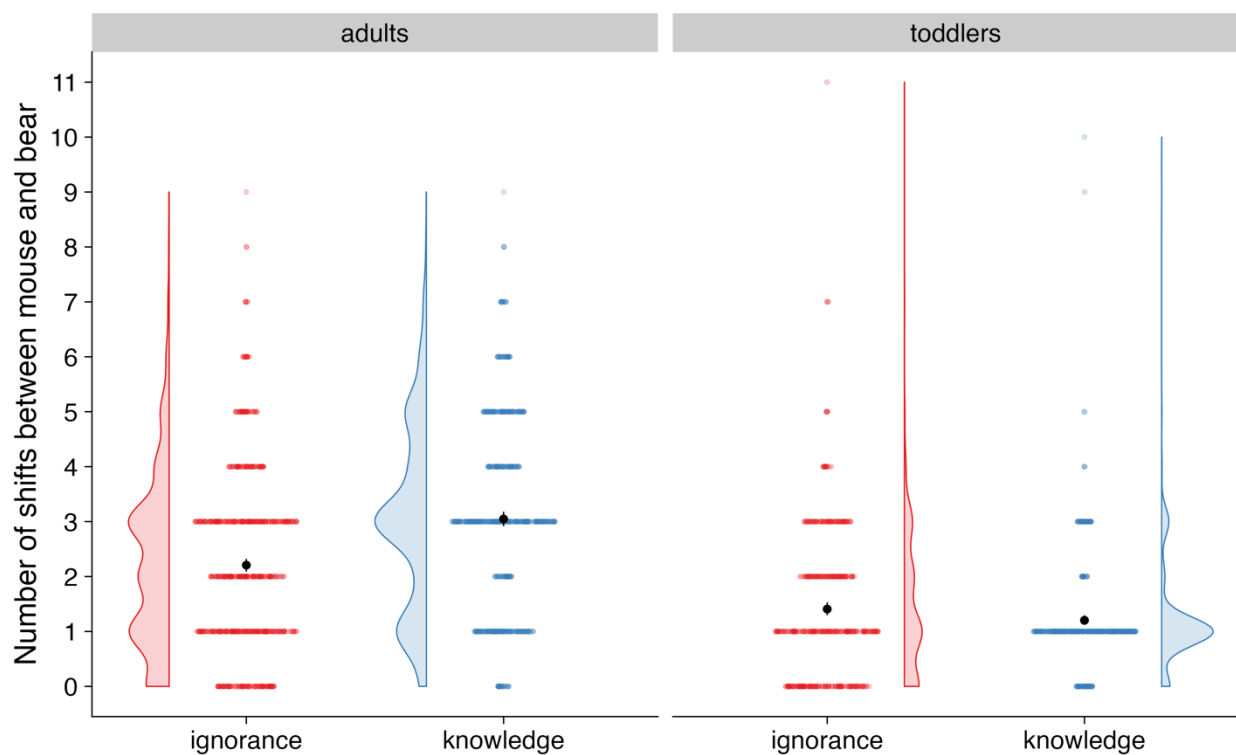
Comparing the number of shifts of toddlers and adults during the location change of the mouse. We fitted a Bayesian mixed-effects model using Poisson family to examine the relation between the number of shifts between mouse and bear and age cohort during location change of the mouse, while accounting for random effects by lab. The effect of condition was negative and approached significance, suggesting a potential reduction in the number of shifts for the ignorance condition compared to the knowledge condition. The main effect of age cohort was positive and credible, Estimate=0.56, indicating that the number of shifts was higher for adults than for toddlers. Importantly, the interaction between condition and age cohort was negative and credible, indicating that the negative effect of condition was more pronounced in the adult cohort (see Figure 8 and Table 6 for the results of Bayesian mixed effects models). Comparing this model to a simpler model without the interaction of condition and age cohort, a Bayes Factor of $BF > 1000$ was computed. This provides strong evidence in favor of including the interaction of condition and age cohort in the model. In order to interpret this interaction, we conducted follow-up analyses separately for toddlers and adults.

For toddlers, there was extreme evidence in favor of the null model compared to the full model that included condition as fixed effect, $BF < 0.01$, suggesting little to no reliable difference in the number of gaze shifts between the knowledge and ignorance condition.

In contrast, for adults, the Bayes Factor of $BF > 1000$ provided extreme evidence in favor of the full model that included condition as fixed effect. In the knowledge condition, adults showed more gaze shifts than in the ignorance condition. These findings demonstrate a robust condition effect in adults but not in toddlers.

Figure 8

Number of shifts between mouse and bear during location change of mouse in the test phase for toddlers and adults in the ignorance and knowledge condition



Note. Error bars represent 95% CIs.

Table 6

Results of the Bayesian mixed effects models for the number of shifts during location change of the mouse

model	term	β	SE	lower CrI95%	upper CrI95%	$\hat{R}$
shifts age cohort	Intercept	0.59	0.02	0.55	0.64	1.00
	Condition	-0.10	0.03	-0.16	-0.04	1.00
	Age Cohort	0.57	0.04	0.48	0.65	1.00
	Condition:Age Cohort	-0.33	0.06	-0.44	-0.22	1.00
shifts toddlers	Intercept	0.26	0.03	0.20	0.32	1.00
	Condition	0.12	0.05	0.02	0.22	1.00
shifts adults	Intercept	0.92	0.03	0.86	0.97	1.00
	Condition	-0.28	0.04	-0.35	-0.20	1.00

AL as a function of number of gaze shifts between mouse and bear during location change. In order to examine the effect of condition and the number of shifts between mouse and bear during location change of the mouse on anticipatory looking, we fitted Bayesian mixed-effects models for each age cohort separately (see Figure 9 and Table 7 for the results of Bayesian mixed effects models). The dependent variable was proportion DLS in the anticipation period. The fixed effects included the main effects of condition, the number of shifts, and their interaction. We also included random intercepts and slopes for number of shifts within each participant and within each lab, allowing us to account for the hierarchical structure of the data and potential variability between labs and participants.

For toddlers, comparing this model to a simpler model without the interaction of condition and number of shifts, a Bayes Factor of $BF < 0.01$ was computed, indicating that the data extremely favored the null model over the full model. A Bayes Factor comparison also provided extreme evidence against including the fixed effect of number of shifts, $BF < 0.01$. These results suggest that, for toddlers, condition influenced proportional DLS, but the number of location

shifts—either alone or in interaction with condition—did not meaningfully contribute to explaining variation in their looking behavior.

For adults, the number of shifts showed a small but credible positive effect, suggesting that more shifts were associated with an increase in proportion DLS. The interaction between condition and the number of shifts was negative and credible, indicating that the effect of number of shifts on proportion looking at test was larger in the knowledge condition than in the ignorance condition. The Bayesian analysis of the interaction effect produced an extremely high Bayes Factor of $BF > 1000$, strongly supporting the presence of this interaction. The effect of the number of shifts also received very strong evidence, $BF = 35.5$, indicating a meaningful contribution of this predictor to the model.

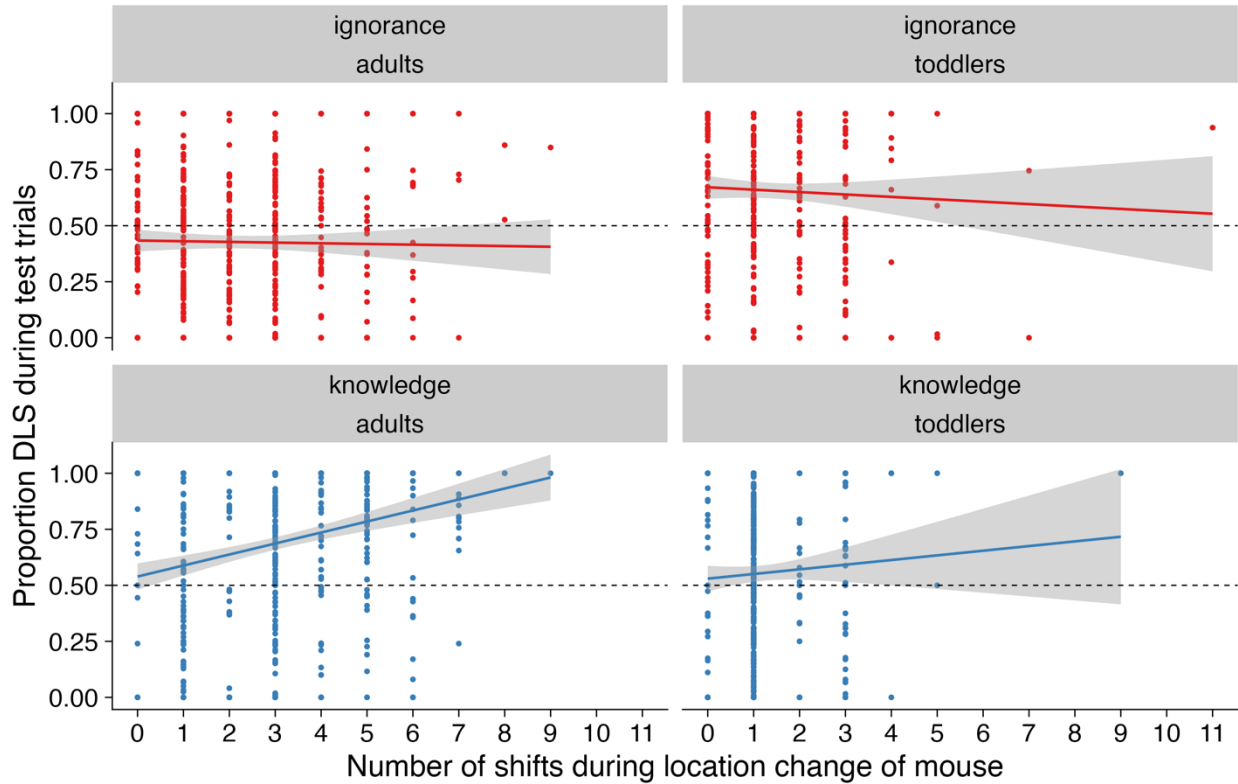
Table 7

Results of the Bayesian mixed effects models for anticipatory looking as a function of the number of shifts during location change of the mouse

model	term	β	SE	lower CrI95%	upper CrI95%	$\hat{R}$
toddlers al shifts	Intercept	0.60	0.02	0.56	0.64	1.00
	Condition	0.12	0.04	0.04	0.19	1.00
	Number Shifts	0.00	0.01	-0.02	0.03	1.00
	Condition: Number Shifts	-0.02	0.02	-0.06	0.03	1.00
adults al shifts	Intercept	0.49	0.02	0.45	0.52	1.00
	Condition	-0.09	0.04	-0.16	-0.02	1.00
	Number Shifts	0.02	0.01	0.01	0.04	1.00
	Condition: Number Shifts	-0.06	0.01	-0.08	-0.03	1.00

Figure 9

AL as a function of number of shifts between mouse and bear during location change of mouse in the test phase for toddlers and adults



Note. Error bars represent 95% CIs.

General Discussion

The overarching aim of the MB2 consortium is to investigate the robustness and replicability of studies showing spontaneous Theory of Mind from infancy across the lifespan. The first project, the initial steps of which are reported here, takes a systematic, sequential bottom-up approach to address anticipatory looking as a measure of spontaneous Theory of Mind and pursues three objectives. First, we aim to develop stimulus material (i.e., videos) that reliably and generally elicits spontaneous, goal-based action anticipation – such that young children and adults look where an agent will go as a function of their goal. In this way, the problem of very

high exclusion rates in previous studies (of children who did not spontaneously anticipate) could be overcome (e.g., Kamps et al., 2020; Kulke, Reiß, et al., 2018; Schuwerk et al., 2018; Southgate et al., 2007). If the first goal can be met, the second goal is to examine whether young children and adults demonstrate action anticipation that is sensitive to basic epistemic states of agents: do they anticipate as a function of whether the agent has or has not seen crucial events and is thus knowledgeable or ignorant of them? If the second goal can be fully met, the third goal, to be addressed in future work, is to test the replicability of the original findings of the false belief/true belief contrast. In this paper, we focused on the first two steps and laid the foundation for future work on the third.

Summary and evaluation of main findings

Concerning our first goal, the present work was successful. In two pilot studies (see Supplement S1) and the main study reported here, toddlers and adults reliably engaged in spontaneous goal-based action anticipation: they looked ahead of time towards the location where an agent would go, given their goal. This licensed the conclusion that the present stimulus material is suitable for studying spontaneous action anticipation and laid the foundation for addressing the second goal in the main study: is spontaneous action anticipation sensitive to the agent's epistemic status?

The results of this study were mixed, even with the large sample size of more than 500 toddlers and 700 adults tested in over 30 labs around the world. By accounting for random effects due to lab, our modeling approach allows us to generalize our findings to toddlers and adults across diverse socio-cultural and lab contexts.

Adults showed clear evidence of the anticipatory looking patterns that one would expect if they engaged in action anticipation that is sensitive to the agent's epistemic status. When the agent (chaser) witnessed the crucial events in the knowledge condition and thus knew where the

target (chasee) was, adults looked in anticipation towards the corresponding location. This anticipation was indicated by both first looks and proportional looking time. When the chaser did not witness the crucial event in the ignorance condition and thus did not know where the chasee was, they did not show such a pattern of anticipatory looking and rather looked at the tunnel exit opposite of the chasee's actual location. Supporting this interpretation, there were clear condition differences in both the first-looks measure and proportion of anticipatory looking.

For toddlers, the findings were different and puzzling. They tended to show qualitatively the same anticipatory looking (both in first looks and in proportional looking time) towards the target location in both knowledge and ignorance trials. The qualitative result in the knowledge condition is consistent with adult behavior. Yet quantitatively, children anticipated substantially more in the ignorance compared to the knowledge condition in their proportion of anticipatory looking – precisely the reverse effect of what one would expect if children tracked the agent's epistemic status.

The overall picture that the present study presents is thus the following: The findings with adults were straightforward and in line with our predictions. In their anticipatory looking, adults engaged in spontaneous goal-based action anticipation (pilot studies and main study), and in doing so, they took into account the agent's epistemic status (main study). Based on this, the next step would be to pursue the third goal: testing whether adults take into account true/false beliefs of an agent in their spontaneous action anticipation. Specifically, adults are expected to anticipate that the agent will go to the actual location of the target in the true belief condition but to the location where the agent falsely believes the target to be in the false belief condition (e.g., Schneider et al., 2011; Senju et al., 2009).

In contrast, the findings with the toddlers are puzzling and not in line with our predictions. Although toddlers did engage in spontaneous goal-based action anticipation (pilot studies and

main study), they did not show clear evidence of taking into account the agent's epistemic status (main study) in the way adults did.

Big open question: How can the puzzling looking patterns in toddlers be explained?

The puzzle is what to make of these findings with children. How did the surprising anticipatory looking pattern of children come about? Why did toddlers not anticipate more clearly the chaser's action in the knowledge condition? And why did they show anticipatory looking to the box with the chasee in the ignorance condition (where they should not do so, or at least to a lesser degree)? A number of initially plausible explanations could be ruled out via exploratory analyses. One such explanation was that behind the grouped data, more nuanced sub-group patterns were hidden. For example, older children could be performing as expected, while younger children were not; or children who anticipated strongly in the familiarization trials could be performing as expected, while children who showed little or no anticipation were not. However, corresponding exploratory analyses along these lines did not find compelling evidence for such sub-group patterns. Neither toddlers' age nor anticipatory looking in the familiarization trials had an effect on the pattern of test trial results. Another explanation was that perseveration from the last familiarization trial to the first test trials (such that children persevere in looking in anticipation to the location they have previously looked to) differentially affected knowledge and ignorance conditions and could thus account for at least parts of the puzzling pattern. But the relevant exploratory control analyses (for details see Supplemental Material) did not find any convincing evidence for such a possibility.

How then can these puzzling anticipatory looking patterns in the knowledge vs. ignorance conditions in toddlers be explained? More specifically, how can we explain why toddlers in the ignorance condition engaged in strong anticipatory looking towards the unpredicted location (where the chasee currently is, unbeknownst to the chaser)? And how can we explain why they

showed only very weak correct action predictions in the knowledge condition – weaker than in the ignorance condition, and weaker than in the familiarization trials? We discuss several possibilities below. These are currently all, needless to say, post hoc speculations. But they may lay the foundation for testing them in future studies.

Timing differences between conditions

One possibility regarding the ignorance condition is that slight differences in timing between the conditions may have posed challenges for toddlers. Specifically, in the ignorance condition, the chaser hides at the back and fails to witness the key events where the chasee moves first to one box and then to another. In contrast, in the knowledge condition, the chaser observes the chasee going to one box, then leaves, returns and witnesses how the chasee moves between the boxes (see Figure 2). As a result, the conditions differed subtly in timing. In the ignorance condition, there was a slightly longer interval between the initial hiding event at location 1 and the anticipatory looking phase. This extended interval may have impaired children's memory of the event, making location 1 less salient and leading them to focus more on location 2 during the anticipation phase. Although the lack of an age effect between 18- and 27-month-olds slightly undermines the memory-capacity explanation (because increasing memory capacity in this age interval should have produced an effect of age), future studies could address this issue by equating the temporal structures of both conditions.

Attentional and other processing demands

A second possible explanation for the obtained pattern of results, and in particular, why toddlers did not anticipate more clearly and strongly in the knowledge condition, may relate to attentional and other processing demands of the knowledge and ignorance conditions.

Regarding attention, the knowledge condition raises particular demands of distributing and coordinating attentional focus. In the familiarization trials, children show clear and strong

goal-based action anticipation. However, in these trials, the chaser remains present in the scene all the time (it never goes towards the back), and the chasee goes to one box in the chaser's presence but does not change to the other box. In the knowledge test trials, in contrast, there is much more going on: the chaser leaves towards the back and then returns, and the chasee first goes to one box, and then relocates to the other. Perhaps dividing attention between the relevant events (chaser is at the back, chasee at the same time in one box) and coordinating it over time (keeping track of what the chaser has witnessed when) was too demanding for toddlers, and as a result they lost track of the narrative structure of the events. Exploratory analyses of gaze shifts between chaser and chasee in toddlers vs. adults may be seen as an indication that there is something to this explanation: Adults seemed to track the chaser's perceptual access, as indicated by many gaze shifts between chaser and chasee during the location change. In contrast, toddlers' attention remained largely on the moving chasee with fewer gaze shifts towards the chaser who was witnessing the relocation in the knowledge condition. Future studies could test more directly whether attentional demands made the present knowledge condition particularly demanding. The chaser could be continuously present all the time, for example, and never leave towards the back (which was introduced to keep the knowledge condition as similar as possible to the ignorance condition) – thus reducing the need to divide and coordinate attention between chaser and chasee. The corresponding ignorance condition could then be realized differently, not such that the chaser leaves, but, for example, such that their view becomes blocked by an occluder, or they fall asleep or are otherwise blindfolded.

Relatedly and more generally, the complexity of the event sequences to be followed and tracked in both conditions may pose excessive performance demands that mask children's competence to understand the agent's epistemic status. Overburdened by such processing demands, toddlers may revert to simpler cognitive strategies influenced by dynamic visual

salience, for example. Future research should aim to address these cognitive constraints by further simplifying task demands and optimizing event timing.

Challenges of understanding the implementation of the ignorance condition

A third possibility is that toddlers' (and adults') anticipation is related to their differing conceptual understanding of the scenes in the ignorance trials. In the ignorance condition, the chaser leaves but its back is visible. This requires monitoring and understanding that this does *not* give the chaser epistemic access to the events, which may overburden toddlers. Additionally, the ignorance condition presents a challenge regarding what to anticipate: should participants expect that the chaser will come out at one of the two exits at random? Or that the chaser will go to the location where the chasee is not? Interestingly, adults' looking behavior in the ignorance condition suggests that they expected the chaser to go to the incorrect location, similar to what might be predicted in a False-Belief scenario or an ignorance-leads-to-error heuristic (e.g., Ruffman, 1996; but see Friedman & Petrashek, 2009). Adults and toddlers looked comparably in knowledge trials with slightly above chance anticipation (albeit stronger in adults), whereas the two groups showed entirely opposite patterns in the ignorance condition. This raises important questions about how toddlers interpreted the ignorance condition. Did they entirely lose track of the chaser's epistemic state? Did they show a 'pull of the real', focusing on the actual location of the chasee? If so, why did this not occur in the knowledge condition? Alternatively, they may have been governed by altogether different assumptions about the events in the scene. This raises the possibility that the ignorance condition may not be the most optimal comparison to the knowledge condition for toddlers. These concerns could be addressed by exploring alternative implementations of the ignorance condition, such as those proposed above.

Differential habituation and task construal across trials between the conditions

A fourth possibility, finally, is that toddlers habituated to and construed the events over time differently in the knowledge and ignorance conditions. In the knowledge condition, from toddlers' perspective the first test trial was the fifth trial (after the four familiarization trials) that was similar in the sense that the chaser went to look for the chasee after watching it hide. So they might have simply begun to lose interest in the task. In contrast, no such habituation may have taken place in the ignorance condition in which the first test trial did differ from the last familiarization trial in that the chaser did not witness the chasee hiding.

In addition, toddlers may have construed the first ignorance test trial in a particularly rich way: For the first time, the chaser did not know where to go — and the chasee knew that. Rather than not keeping track of the agents' perspectives on the scene (under-thinking), toddlers may have engaged in very complex reasoning (over-thinking) about such perspectives along the following lines: They may have looked toward the chasee's box because they expected the chasee to signal its hiding location in some way, to help the chaser find it and allow their cooperative game to continue. From this perspective, toddlers thus not only understood the chaser's ignorance, but they expected the chasee to understand it too and to act accordingly. This made the chasee, rather than the chaser, the focus of toddlers' anticipation: They anticipated that the chasee would somehow signal its location, to help the chaser find it. This interpretation of the ignorance condition also indicates that it may measure something different in toddlers and adults and that anticipatory looking in toddlers can have a different meaning than it has in adults — i.e., a lack of measurement invariance (Meredith, 1964). While such a very rich interpretation does not receive any direct support from the present data, it could be put to systematic test in future studies (for example, by having the two agents interact in less cooperative ways).

Conclusion and future directions

The current large-scale study examined the robustness and reliability of studies using anticipatory looking as a measure of spontaneous Theory of Mind in toddlers and adults. The novel stimuli designed for this study reliably elicited spontaneous goal-based action anticipation, as shown in two pilot studies and in the main study. Spontaneous anticipatory looking occurred to a higher degree and resulted in lower exclusion rates than in previous studies - confirming the importance of the baseline checks, especially in replication studies. In this sense, the current study provides robust stimulus material suitable to elicit goal-based AL in both toddlers and adults.

The main study tested whether toddlers and adults, in their spontaneous action anticipation, take into account the epistemic status of an agent who witnesses relevant events and knows where the target is (knowledge condition) or fails to do so (ignorance condition). Adults clearly did take into account the agent's epistemic status and distinguished between the knowledge and the ignorance condition: they anticipated that the agent would go to the target in the knowledge condition, but expected the opposite in the ignorance condition. In contrast, toddlers showed anticipatory looking to the target location in both conditions, but did so in quantitatively stronger ways in the ignorance than in the knowledge condition, was the opposite than expected.

Future research with adults could build on the present findings, for example, by testing whether adults engage in belief-based action anticipation and thus whether original findings of implicit Theory of Mind in adults can be replicated. Future research with children, in contrast, should first sort out the sources of the unexpected results found here. This will require systematic follow-up studies along several lines: First, the knowledge-ignorance condition contrast could be implemented in alternative, ideally simpler, ways as suggested in the above discussion. Second, an interesting extension would be to test whether young children do indicate some understanding

of the present knowledge-ignorance contrasts in other measures. Anticipatory looking itself may be a demanding measure due to its predictive nature (cf. Johnson et al., 1991). Postdictive measures, in contrast, such as looking behavior and pupil dilation in response to events that are/are not expected given the agent's knowledge/ignorance may be more sensitive to uncover early competence (e.g., Daum et al., 2012). Currently, a spin-off project (<https://manybabies.org/MB2P/>) is running the first follow-up studies in this direction. Finally, looking at children at different ages, for example, older children approaching an age of verbal Theory of Mind reasoning, could shed light on whether their behavior in this task may be related to an underlying conceptual understanding (e.g., Grosse Wiesmann et al., 2018). Taken together, these future studies will hopefully shed more light on the reality and robustness of implicit Theory of Mind from infancy to adulthood.

To conclude, this study represents a critical step forward in understanding the development and robustness of spontaneous Theory of Mind across the lifespan. By developing novel, reliable stimuli and implementing a large-scale, multi-lab approach, it has laid the groundwork for replicable research in this domain. The findings demonstrate that adults' anticipatory looking aligns with epistemic sensitivity, while the unexpected, puzzling patterns in toddlers challenge existing assumptions and thus open up exciting new avenues for future research. By addressing these puzzles, this work paves the way for deeper insights into the developmental trajectory of Theory of Mind and the cognitive mechanisms underlying its expression in infancy, childhood, and beyond.

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Supplemental Material

This document contains supplemental material of the manuscript:

Schuwerk, T.*, & Kampis, D.*, et al. (2025). Action anticipation based on an agent's epistemic state in toddlers and adults. [Manuscript submitted for publication] (*shared co-first authorship).

S1. Pilot Studies

The familiarization trials were developed to convey information that is necessary for correct action predictions in this paradigm. First, the agent's goal is introduced, i.e. the chaser wants to catch their partner (the chasee). Second, the situational constraints of the scene are shown. A barrier (fence) divides the scene so that the other side can only be reached by going through a y-shaped tunnel. Yet, it had to be clear that the fence is not a visual barrier, meaning that the chaser can see everything that takes place on the other side. Third, the familiarization trials should teach the timing of events, particularly, how much time the chaser spends in the tunnel and when their reappearance is to be expected. We piloted the stimuli with adults and toddlers between 18 and 27 months of age, the core age range of our main study. All analysis scripts can be found on GitHub (<https://github.com/manybabies/mb2-analysis>).

Pilot 1

In the first pilot study, we wanted to get an estimate of the level of correct goal-based action predictions with these novel stimuli. We presented a total of eight familiarization trials. An observation of changes in the anticipation rate over trials would help us to determine the optimal number of familiarization trials. Further, we used this pilot to test the general procedure (i.e., data collection in different labs, preprocessing and analysis of raw gaze data from different eye-

trackers). We also checked whether gaze patterns indicated any issues with perceptual properties of stimuli, such as distracting visual saliencies. Data for this pilot study was collected between February and July 2019.

Methods

Participants. Seven labs¹ tested a total of 65 healthy full-term toddlers (28 males; Mean age = 23.14 months; range: 18.25 to 26.84 months). Data from eight additional toddlers were excluded from the analyses. Three did not complete the full experiment, another three did not complete at least six trials. Two toddlers had to be excluded due to technical problems with data collection (e.g., calibration of eye-tracker). At the trial level, four additional trials were excluded because the trial data was incomplete (as determined by not having at least 32 s of eye-tracking data for that trial, from the beginning to the end of the trial). A total of 42 adults were tested in three labs [5 males, 1 male/other, 1 N/C (not collected); Mean age = 24.10 years; range: 19 to 53 years]. One adult was excluded because this participant did not complete at least six trials. We asked contributing labs for a minimum sample size of 3-5 participants per age group. We reasoned that the resulting minimum total sample of 27-45 participants per age group would be large enough for an initial estimate of anticipatory looking (AL) behavior. The contributing labs were independently responsible for obtaining informed written consent and reimbursing participants. Each lab acquired ethics approval. Central data analyses only used de-identified data. Video recordings of participants were archived locally at each lab following the local data protection regulations.

¹ The contributing labs were: CEU Cog Dev Center, Central European University, Budapest; Babylab Copenhagen, University of Copenhagen, Denmark; Göttinger Kindsköpfe, Georg-August-Universität Göttingen, Germany; LMU Babylab, Ludwig-Maximilians-Universität München, Germany; Babylab Uni Trento, University of Trento, Italy; Center for Infant Cognition, University of British Columbia, Canada; Infant Learning and Development Lab, University of Chicago, USA

Task and Procedure. Toddlers were tested in a quiet room of nurseries or laboratories, after their caregivers read and signed the informed consent form. They sat on an educator/caregiver's lap or on a car seat, centered in front of the monitor used to display the stimuli at a distance of about 60-80 cm. Educators or caregivers were instructed to remain silent and to wear black glasses or close their eyes to avoid erroneous tracking of their eyes. The experimenter was behind a curtain/room divider and controlled stimulus presentation. Depending on the lab setup, the following eye-tracking systems were used: Tobii T60 (two labs), Tobii T120 (two labs), EyeLink 1000 Plus (two labs), SMI250Redmobile (two labs), SMI iView X Hi-Speed 1250 (one lab). For each lab the following information was collected: type of eye-tracker apparatus, trial order condition (A or B), any procedural or technical error that occurred during the experimental session, location of the lab they were tested in (laboratory or nursery).

The task consisted of a calibration check, eight familiarization trials and another final calibration check. After an initial attention getter, participants were presented with the calibration check that consisted of an animated star with sound, moving and stopping at four locations. The familiarization trials were as described in the Methods section of the main study, with the following deviations: In the upper part of the tunnel there was a small window that allowed participants to watch the agents moving inside the upper part of the tunnel before it forked. Further, unlike in the final familiarization trial version, a chime sounded at the moment the chaser disappeared from the tunnel window, indicating the start of the anticipatory period. The starting location of the chasee (left or right half of the upper part of the scene) and the box the chasee ended up (left or right box) were counterbalanced, resulting in a total of four familiarization trial versions [started from the right and ended up in right box (RR); started from the right and ended up in left box (RL); started from the left and ended up in right box (LR); started from the left and ended up in left box (LL)]. Each of these versions was presented twice in two pseudo-randomized

orders (Order A: LL1, LR2, RR2, RR1, LL2, RL2, LR1, RL1; Order B: RL1 LR1, RL2, LL2, RR1, RR2, LR2, LL1). Half of the participants in each lab group were randomly assigned to one of the two orders.

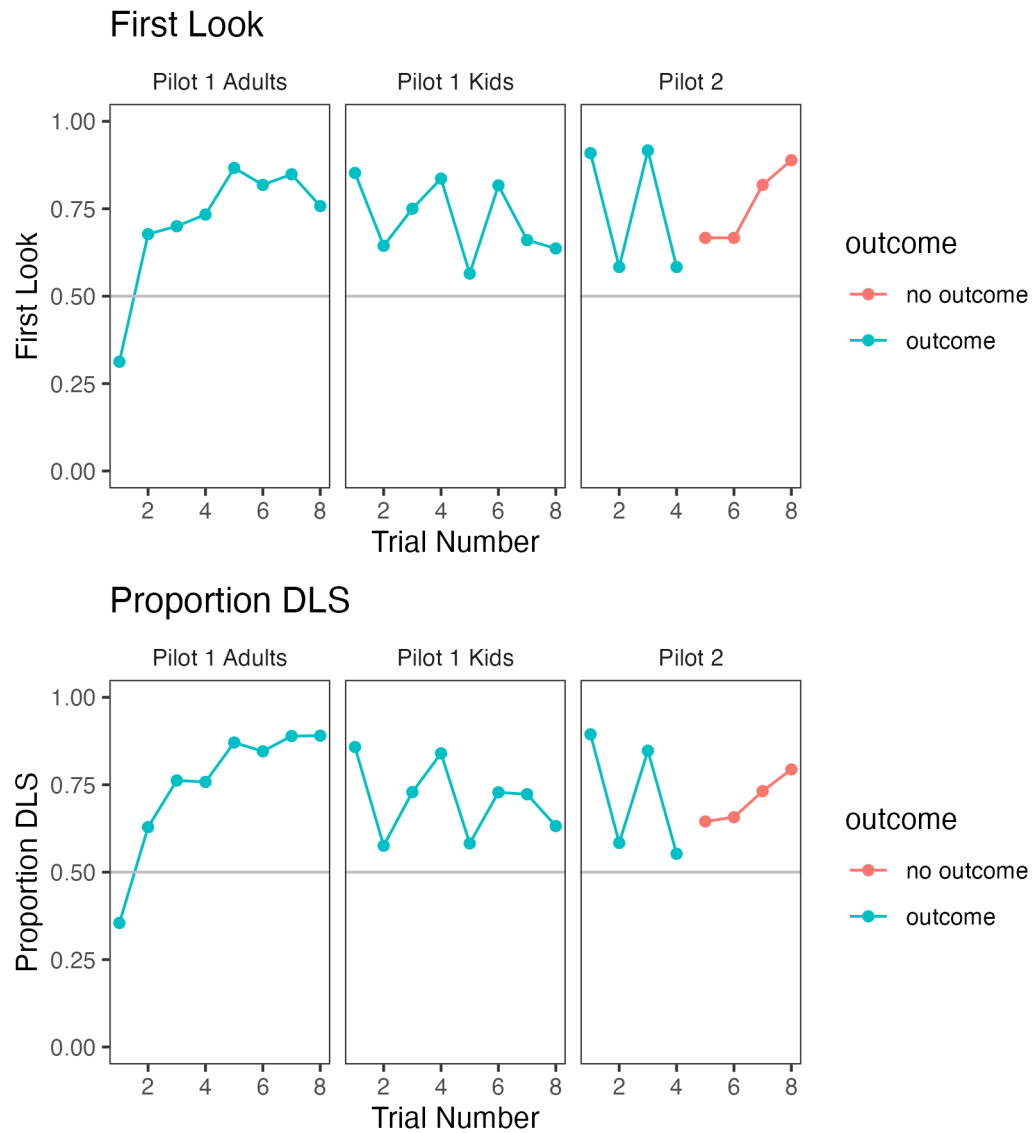
Data Analysis. The labs exported the raw gaze data in the format the respective eye-tracking software allowed. The participants' demographic information and details about the test session were collected in standardized spreadsheets. Each lab provided the raw gaze data and de-identified demographic information with Google Drive. Data preprocessing was identical to the procedure of the current study. For details refer to the Methods section of the main manuscript.

Results

Descriptive Statistics. In Figure S1, we show the toddlers' proportion of first looks and the proportion differential looking score (DLS) during the anticipatory period of each trial. Figure S2 (plots labeled pilot 1) shows the proportion of looking of toddlers and adults as a smooth curve, generated by binning the data and averaging the proportion looking at each time point across all participants. We saw robust evidence for looks to the target relative to the distractor during the anticipation period, as evidenced by the red lines being consistently higher than the blue lines. In Figure S2, we separated trials into two blocks (Trials 1-4 and Trials 5-8). For toddlers in Pilot 1, we see similar rates of anticipation for Trials 1-4, as in Trials 5-8. In fact, anticipation is slightly lower in Trials 5-8 than in Trials 1-4. For adults, we see an increase in the anticipation rate in Trials 5-8. The heatmaps in Figure S3 illustrate the distribution of looks to scene locations during the anticipatory period. We found that a large proportion of anticipatory looks was directed to the tunnel exits. Substantially fewer looks fell onto the boxes. Unexpectedly, many looks were attracted by the tunnel window (the location where the chaser was last seen).

Figure S1

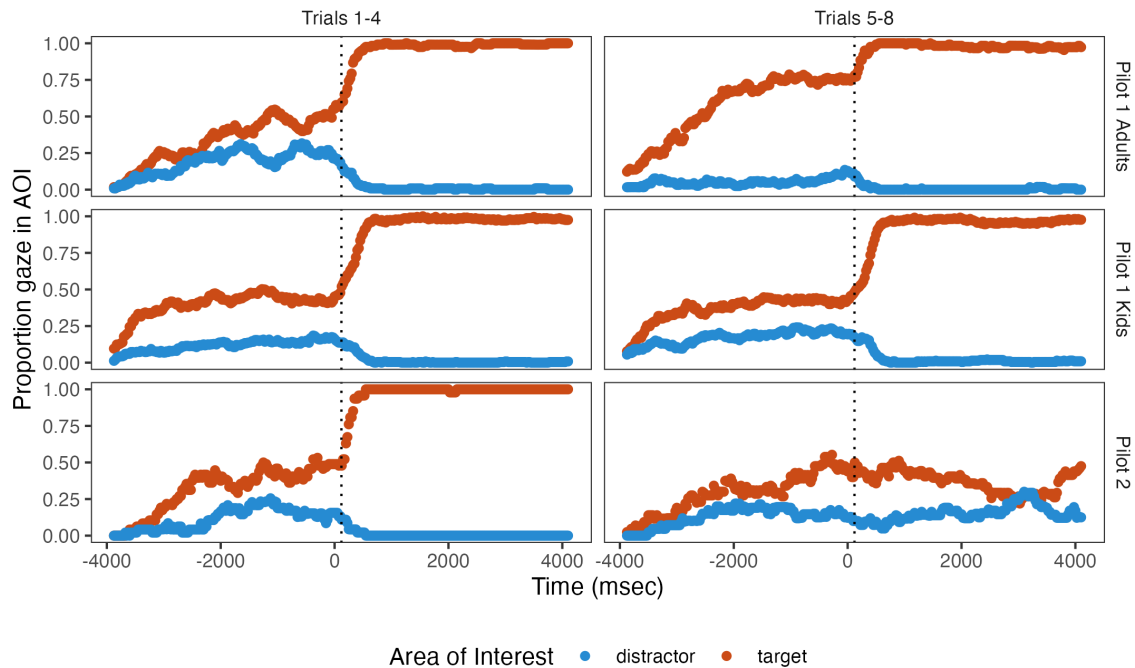
First looks and proportion DLS from Pilot 1 and Pilot 2 for each trial



Note. Top: Proportion of first looks to the target as a function of trial number; Bottom: Proportion DLS as a function of trial number.

Figure S2

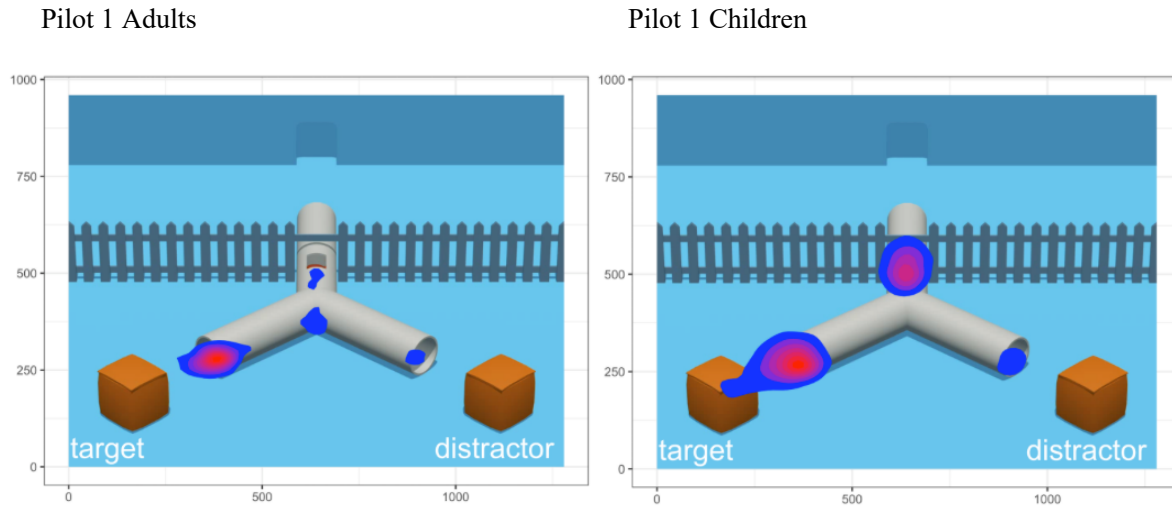
Binned proportion looking, averaged across all participants and trials



Note. The left column comes from Trials 1-4, the right column from Trials 5-8. The vertical dotted line represents the disambiguation time. The red points represent looks to the target, the blue points represent looks to the distractor. The lower right panel shows data of the no outcome trials of Pilot 2. In these trials, the chaser never reappeared from the tunnel and thus there is no disambiguation between target and distractor because no agents reappear that could be fixated.

Figure S3

Heatmaps showing the distribution of looks in the anticipatory period in Pilot Study 1 (left: adults; right: toddlers)



Note. Larger and reddisher areas indicate a greater amount of looks toward the respective location.

Inferential Statistics. To further assess the pilot data and test our proposed analysis described in the main text, we ran two Bayesian mixed effect models as described in the main manuscript, the first using first look location as the dependent variable and the second using proportional looking score as the dependent variable. For the first look analysis, we defined the first look location as in the main text (corresponding roughly to the first look of 150 ms or more in the same AOI). We calculated the proportion DLS to the correct AOI during the full 4000 ms anticipatory window by $\text{correct AOI looks} / (\text{correct AOI looks} + \text{incorrect AOI looks})$, excluding looks outside of either AOI. The anticipatory eye movement window was defined 120 ms after the first frame when the chaser had completely entered the tunnel and 120 ms after the chaser reappeared from the tunnel.

Because we wanted to ask if participants were attentive and could still make predictive looks at the end of the familiarization phase, we coded the trial number such that the last trial during the familiarization phase (the 8th in pilot 1) is set to 0, with trials 1 through 7 are coded as -7 to -1, respectively. We used the priors described in the main text of our analysis plan. Our base model was as follows, where measure refers to the dependent variable (either first look or the proportional looking score):

$$\text{Measure} \sim 1 + \text{trial_number} + (\text{trial_number}|\text{lab}) + (\text{trial_number}|\text{participant}).$$

We fitted a reduced model for model comparison:

$$\text{Measure} \sim 0 + \text{trial_number} + (\text{trial_number}|\text{lab}) + (\text{trial_number}|\text{participant}).$$

We then calculated the Bayes factor, which we interpret as described in the main text.

Toddlers. For the first look analysis, the intercept estimate was .44, (CrI95% = 0.07, 0.80). This corresponds to a point estimate of a 61% probability of the first look to be mapped onto the target as opposed to the distractor. The Bayes factor comparing the model with and without the intercept was 1.52, which is inconclusive by our criteria (Schönbrodt & Wagenmakers, 2018). For the proportional looking score main model, the model estimate for the intercept was 0.16 (CrI95% = 0.10, 0.23). This can be interpreted as a point estimate of a 66% probability of looking at the target. The Bayes factor was 493.25, which was strong evidence in favor of the full model and which strongly suggests that toddlers looked more towards the target than towards the distractor during the anticipation period.

Adults. For the first look analysis, we obtained a model estimate of 1.95 (CrI95% = 1.42, 2.48). This corresponds to a probability of 88% that the first look is to the target. The Bayes factor was > 1000, which was evidence in favor of the full model. For the proportion DLS

analysis², the Bayes factor was also > 1000 , which was evidence in favor of the full model. This suggested that adults had a higher proportion of looking at the target than chance level. The model estimate for the intercept was 0.46 (CrI95% = 0.38, 0.54). Based on these analyses, it is clear that adults looked more to the target than the toddlers did, and it appears this was driven by Trials 5-8, as can be seen in Figure S2. Adults learn to anticipate the target and, on later trials, very rarely look at the distractor.

Discussion

Based on the first pilot, we drew the following conclusions: (1) Toddlers and adults show anticipation during the anticipatory period, and thus the paradigm seems successful at eliciting anticipation. (2) Over the course of eight trials, toddlers and adults remained attentive and showed anticipatory behavior even during the last trial of the familiarization phase. (3) Four familiarization trials seem to be sufficient and there do not appear to be strong additional benefits of running additional trials. Crucially, trials five to eight did not help to increase the overall anticipation rate for toddlers, as shown in Figure S2. Note that in the adults' sample AL slightly increased after trial 4. We nonetheless decided to use four familiarization trials in the main study because we reasoned that it is more important to avoid fatigue or boredom in the toddlers' sample than to get even higher anticipation rates for adults.

It is important to note that our decision to include 4 familiarization trials was based on (1) conceptual and practical methodological considerations also considering previous studies and (2) the pilot study results. Replication studies of Southgate et al. (2007) pointed to issues with the familiarization phase and that the two trials of the original study might not be enough to familiarize toddlers with the scenario (e.g., Kamps & Southgate, 2020; Schuwerk et al., 2018).

² We note that the base model for the proportion DLS analysis in adults had divergent issues. These issues were not resolved after adjusting the alpha level to a very high number (e.g., 0.999999). Thus, the results needed to be interpreted with caveat.

On the other hand, to avoid unnecessarily increasing the overall length of the task and to prevent poor anticipatory looking due to fatigue or boredom, we did not want to include too many familiarization trials. In the discussions preceding the pilot data analysis, we came to the conclusion that four trials reflect such an optimal trade-off. The pilot data results of the toddlers then supported this decision insofar as we observed a looking bias towards the correct location already in trials 1-4, without additional benefit of trials 5-8. Due to the exploratory nature of the pilot studies, we refrained from running inferential statistics in addition to the visual inspection of the first look and proportion DLS data, as well as of the time series illustration, which all converged on this interpretation (see supplementary Figure S1 and S2).

The duration of the anticipatory period was set based on durations used in previous studies. Earlier studies found action outcome-contingent anticipatory looking with anticipatory phases ranging between approximately 2-3.5 seconds (Low & Watts, 2013; Meristo et al., 2012; Surian & Geraci, 2012; Thoermer et al., 2012). To make sure we were not losing anticipatory looks by cutting off too early, we decided to use a time period of 4 seconds. The pilot data showed no evidence for a decline in anticipatory looking towards the end of the anticipatory period (see time series plot in Figure S2), which supported this decision.

Further, the distribution of looks in the anticipatory period helped us to evaluate the appropriateness of our AOI dimension, in particular whether restricting AOIs to the tunnel exits not including the adjacent box optimally captures goal-directed anticipatory looks. By increasing the AOI dimensions so that they cover both the tunnel exit and the box, we could potentially detect more goal-directed anticipatory looks. On the other hand, looks to the box cannot unambiguously be interpreted as anticipations of the chaser's upcoming action. Participants might look to the box simply because this is where the chatee is, anticipating that the chatee might jump out of the box again. Thus, we concluded that restricting our AOIs to the tunnel exits

–the location where the chaser will reappear– is the more conservative and more unambiguously interpretable measure of goal-directed action prediction. The result of our pilot study corroborated this strategy. The larger proportion of anticipatory looks was indeed directed to the tunnel exits and not to the boxes. Based on this finding, we concluded that using the tunnel exit AOIs is the sharper measure of goal-directed action predictions without a substantial loss of looks that could also reflect action predictions but are directed elsewhere (e.g., to the box).

An unexpected result of Pilot 1 was that during the anticipatory period, many fixations were attracted by the tunnel window where the agent was last seen. This was potentially problematic since looking at the window could lead to a reduced amount of anticipatory looks to the target/distractor AOIs. Initially, the window was added to the tunnel with the aim to increase AL (cf., Surian & Franchin, 2020). But the results suggested that it may have been distracting, and so we removed the window for Pilot 2.

Pilot 2

To further hone our stimulus design, we conducted a second pilot. First, we removed the potentially distracting tunnel window from all trials in Pilot 2. Second, we tested another method to increase AL. We asked whether a chime as an arbitrary timing cue helps to elicit AL to the tunnel exits in (future) test trials in which the agent does not reappear at one of the tunnel exits (because these test trials stop after the end of the anticipatory phase without showing the agent's action outcome). To this end, we presented the first four familiarization trials showing the outcome associated with the chime, i.e., the chime announced the reappearance of the chaser, and four subsequent familiarization trials without an outcome, i.e., the chime sounded, but the chaser did not reappear. We reasoned that if participants learn in the first four trials that the chime indicates the chaser's reappearance, we should see an increase in AL right after the chime

sounded. Further, this increase should also be observable in the last four trials in which the chaser does not reappear. Data collection for this pilot started in January 2020 and had to stop due to Covid-19 outbreak in March 2020.

Methods

Participants. A total of 12 healthy full-term toddlers participated in the second pilot study (6 males; Mean age = 24.15 months; range: 19.14 months to 26.05 months). One additional toddler was tested but excluded from the analyses because this toddler did not complete at least six trials. An additional one trial was excluded as the toddler did not look at least 32 seconds during this trial. We asked five labs³ to contribute a minimal sample size of four toddlers. Yet, data collection had to stop due to the Covid-19 outbreak.

Task and Procedure. The task and procedure were similar to Pilot 1. In this study, the following eye-tracking systems were used: Tobii T60 (one lab), Tobii T120 (one lab), EyeLink 1000 Plus (two labs), and Tobii Pro Spectrum (one lab). After the initial attention getter, participants were presented with the calibration check as in Pilot 1, eight familiarization trials and at the end, again the calibration check. The familiarization trials started by showing the same scene as in pilot 1, except that the window was removed from the tunnel. The trials differed in whether they displayed an outcome (i.e., the chaser exits the tunnel and the two agents rejoin) or not (i.e., trial stopped after the anticipatory period). The first four trials showed the outcome, the last four trials did not. Unlike in the first pilot, the chime now sounded the moment the chaser reappeared at one of the tunnel exits in the outcome trials. In the no outcome trials, the chime sounded the same moment, yet now the chaser did not appear. Again, the trials were presented in two pseudo-randomized orders [Order A: outcome (LR, LL, RR, RL), no outcome (LL, RL, LR,

³ The contributing labs were: CEU Cog Dev Center, Central European University, Budapest; Babylab Copenhagen, University of Copenhagen, Denmark; Infant Learning and Development Lab, University of Chicago, USA; Leiden Babylab, Leiden University, The Netherlands; Brigham Young University, USA.

RR); Order B: outcome (RL, RR, LL, LR), no outcome (RR, LR, RL, LL)]. Half of the participants in each lab group were randomly assigned to one of two orders.

Data Analysis. Data preprocessing was analogous to Pilot 1.

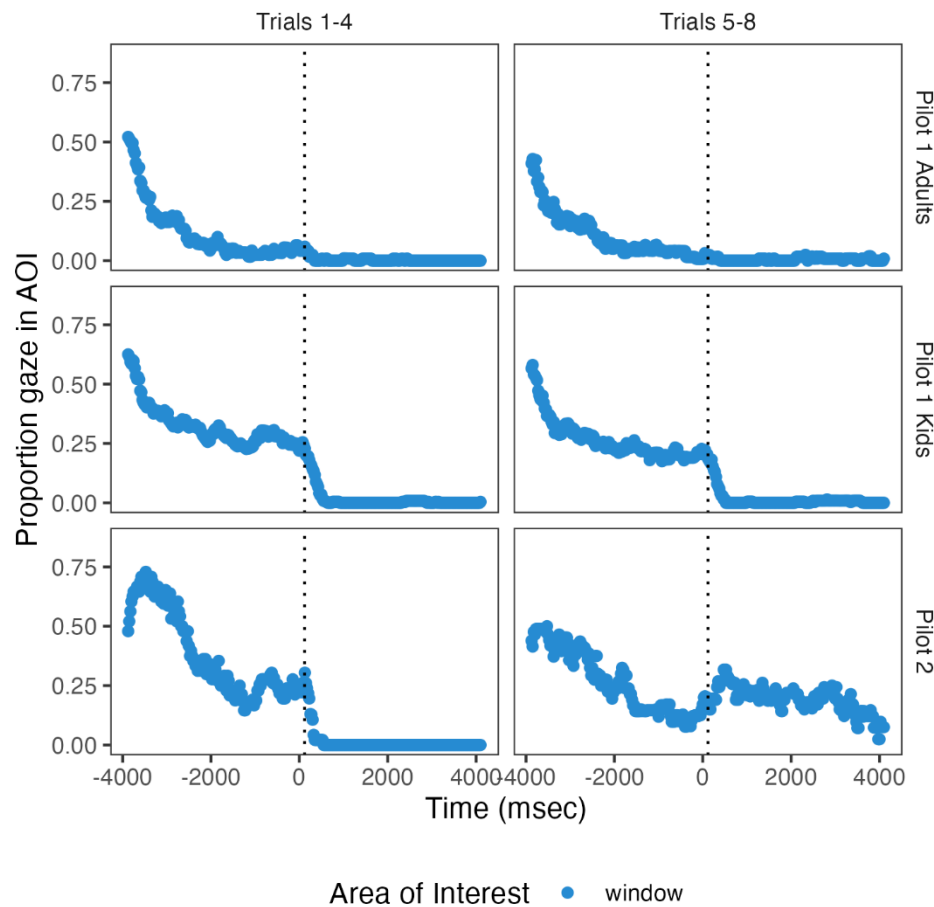
Results and Discussion

As can be seen in Figures S1 and S2, we found a similar pattern of results in both conditions of Pilot 2 (with outcome and without come) as we did in Pilot 1. We saw more looks directed towards the target than to the distractor. As described above, all trials in Pilot 2 lacked the tunnel window, whereas all trials in Pilot 1 included the tunnel window. Thus, we can assess the effect of the tunnel window by comparing Pilot 2 to Pilot 1. We found that the removal of the tunnel window did not appear to increase or decrease AL in Pilot 2 in any clear way. In fact, even after the removal of the window, a substantial amount of gaze was attracted towards the location where the window had been in Pilot 1 (for an illustration, see Figure S4). An explanation for this pattern of results is that not the window itself but its location in the center of the scene attracted visual attention. Previous research documented a central fixation bias in infants, toddlers and adults when viewing complex visual scenes (Tatler, 2007; van Renswoude et al., 2019).

By comparing the outcome and no outcome conditions in Pilot 2, we were able to assess whether the use of the chime helps AL. We did not find evidence that the chime helped to increase AL, and the majority of anticipatory looks to the tunnel exits happened before the chime sounded. As with Pilot 1, we ran a series of Bayesian mixed effect models to quantitatively evaluate anticipation. As we had a much smaller sample in Pilot 2, our Bayesian analyses were broadly inconclusive and did not favor either the full or null model. (Bayes factors fell between 0.1 and 3), suggesting that we did not have sufficient data to conclude whether the evidence is in favor of the full model or the simpler model. But, by comparing the results to the results of Pilot 1, we are confident that the results of Pilot 2 are qualitatively similar.

Figure S4

Proportion looks to the area where the window is (in Pilot 1) or would be if it were there (in Pilot 2) across conditions



Note. These graphs show that, at the time that the chaser disappears at around -4000 ms, there are many looks to the window/center of the screen. Over the course of the anticipation period, as more participants look to the target and distractor, there are fewer looks to the window. At the time of disambiguation, which occurs in all panels except for Pilot 2, Trials 5-8 (the no outcome condition), any remaining looks to the window disappear.

Conclusions

In both pilot studies, we found that participants produced goal-directed action predictions. The combined analysis using AOIs around the tunnel exits revealed a looking bias towards the exit at which the chaser reappeared following their goal to catch the chasee. We are thus confident that participants clearly predicted the agent's action and did not just look at the chasee's location, anticipating something else. The changes of stimulus features in Pilot 2 did not affect AL rates. To reduce the complexity of the stimuli, we decided to use the stimuli without the tunnel window. Further, we removed the chime from the final version. In sum, we conclude that these novel stimuli sufficiently elicit goal-directed action predictions and are thus suited to serve as familiarization trials in the study described in the main text.

S2. Notes on and deviations from the Stage 1 manuscript and preregistered methods and analyses

Below, we have compiled a list of notes on and deviations from the Stage 1 manuscript and the preregistered methods and analyses available at <https://osf.io/7ubkw>.

- Figure 3: We moved Figure 3 from the main manuscript to the Supplement.
- Counterbalancing table: The original counterbalancing table was extended.
- Familiarization Trial Order: In our study, one deviation from the preregistration involved the familiarization trial order. Originally, familiarization order A was specified as *Fam_LR*, *Fam_RR*, *Fam_LL*, *Fam_RL*, and order B as *Fam_RL*, *Fam_LL*, *Fam_LR*, *Fam_RR*. However, in the final implementation, the order was adjusted such that familiarization order A followed the sequence *Fam_LR*, *Fam_LL*, *Fam_RL*, *Fam_RR*, while familiarization order B was *Fam_RL*, *Fam_RR*, *Fam_LR*, *Fam_LL*.
- Online Gaze Recording: We originally anticipated to allow for flexibility in gaze coding methods, including manual frame-by-frame annotation from video recordings and the use of various virtual data collection platforms. However, to maintain consistency in data collection across labs, we ultimately implemented a uniform approach in which all labs used the MB-ManyWebcams WebGazer setup (Steffan & Zimmer et al., 2023). This deviation ensured greater standardization in eye-tracking procedures and minimized methodological variability across sites.

S3. Further Supplemental Information: Methods

Overview on employed eye-tracking systems

Table S1 provides an overview of the different eye-tracking systems used by participating labs.

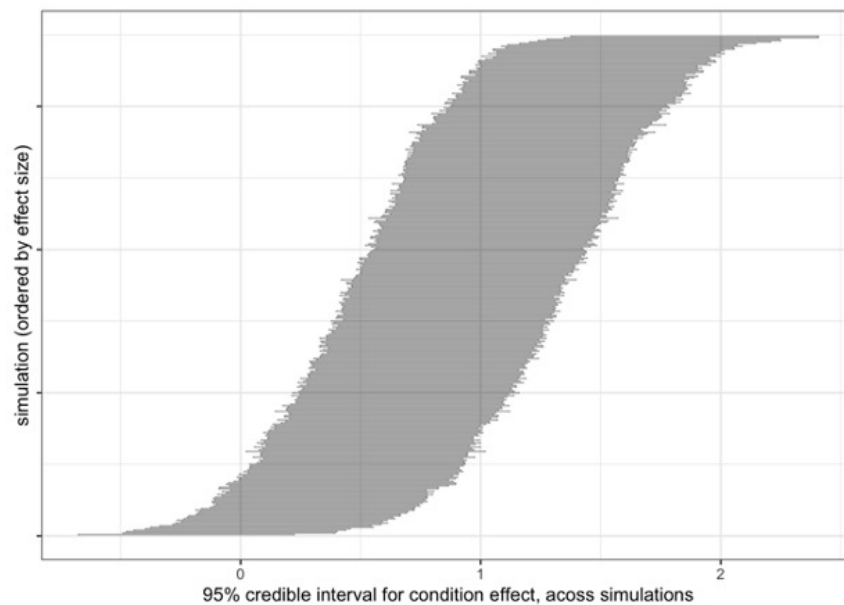
Table S1

Overview of eye-tracking systems, software type, and sampling rates used

Eye-tracking system	N	Software	Sampling rate
EyeLink 1000	1	EyeLink Experiment Builder	500 Hz
EyeLink 1000 Plus	13	EyeLink Experiment Builder	1000 Hz, 250 Hz, 500 Hz
EyeLink Portable Duo	1	EyeLink Experiment Builder	1000 Hz
SMI RED250Mobile	1	SMI Experiment Center	60 Hz
SMI Redn	2	SMI Experiment Center	60 Hz
Tobii Pro Fusion 120 Hz	8	Tobii Pro Lab, PsychoPy	120 Hz
Tobii Pro Fusion 120 Hz/Tobii X3-120	1	Tobii Pro Lab	120 Hz
Tobii Pro Spectrum	14	Tobii Pro Lab, PsychoPy	1200 Hz, 60 Hz, 120 Hz, 300 Hz
Tobii Pro X3-120	2	Tobii Studio	120 Hz
Tobii T120	2	Tobii Studio, Tobii Pro Lab	120 Hz
Tobii T60	2	Tobii Studio	60 Hz
Tobii TX300	8	Tobii Pro Lab, Tobii Studio	300 Hz
Tobii X120	7	Tobii Studio, Tobii Pro Lab	120 Hz
Tobii X3-120	1	Tobii Pro Lab	120 Hz

Design Analysis

We implemented a simulation-based design analysis to demonstrate the range of BFs we might expect to see, given a plausible range of effect sizes and parameters. We focussed this analysis on our key analysis of the test trials (as specified below), namely the difference in AL on the first test trial that participants saw. Figure S5 depicts the effect sizes of simulated experiments.

Figure S5*Effect sizes of simulated experiments*

Note. Ordered by effect size (from left to right), 95% credible intervals for the key effect (in logit space) for our simulated experiments that use first look as the dependent variable.

Questionnaires and test session information

Using a questionnaire (filled out during the lab session or online for remote testing procedures), we collected the following demographic information from the participating toddlers: gender, chronological age in days, nationality of the toddler, estimated proportion of language exposure, preterm/full-term status, current visual or hearing impairments, any known developmental concerns, information about siblings (number, gender, age), duration of time the toddler spends with caregivers and in day-care. From their caregivers the following information was collected: gender, nationality, native language(s), level of education. For the adult sample,

the following demographic information was collected: gender, chronological age in years, and level of education.

Additionally, we collected the following information for each participant: name of lab the participant was tested in, academic status of the experimenter involved in the test session (e.g., volunteer, undergraduate, graduate, post-doctoral, professor), the type of eye-tracking apparatus used including sampling rate and screen dimensions (for eye-tracking procedures), date of testing, trial order condition the participant was assigned to, any procedural or technical error that occurred during the session and further reasons for exclusion, and the type of recruitment method the lab used. For the toddlers' sample, we additionally asked for the amount of experience the experimenter has in testing toddlers, and whether the toddler sat on the caregiver's lap or in a seat. The requested demographic information that was not used in the registered confirmatory and/or exploratory analyses of this study was collected for further potential follow-up analyses in spin-off projects within the MB framework.

Stimuli

General Scene Setup

The depicted scene comprised an open space colored in blue. A horizontal picket fence divided the space into two sections (upper: approx. one third; lower: approx. two thirds). In the upper section, initially two animated, same-sized agents were seen: a brown bear (chaser) and a yellow mouse (chasee). The agents communicated using pseudo utterances. When they moved, footsteps could be heard. The back of the upper section was formed by a wall with a small, central door through which the agents could enter and leave the scenario. Leaving through this door partially covered the agent, with the lower part of the body still visible. In the lower section of the scene, two identical brown boxes with moveable lids were located (one on the left and one

on the right side). A white, centrally located, inverted Y-shaped tunnel connected both sides of the fence. One entrance was located in the upper section, while two identical exits were located in the lower section. Each exit in the lower section pointed towards the left or right box, respectively. The agents could move from the upper to the lower section of the scene by walking through the tunnel.

Familiarization Trials

All participants viewed four familiarization trials. Each trial started with the chaser and the chasee playing tag in the upper section of the scene. That is, the chasee ran off in a circle and was closely followed by the chaser (~4 s). When the chasee stopped, the chaser caught up and they did a high five (~1 s). After separating again, the agents stood next to each other in front of the tunnel's entrance (left or right position counterbalanced) (~3 s). Next, the chasee made eye contact with the chaser (~2 s) and left for the tunnel. The chaser watched closely as the chasee walked towards the tunnel and entered it (~2 s). The chaser then positioned themselves centrally in front of the tunnel entrance (~4 s). While the chasee was walking through the tunnel for four seconds, there was a sound of footsteps. The footsteps ceased when the chasee left the tunnel through one of the two exits (left or right, counterbalanced) in the lower section (~3 s). At this point, the chasee briefly stopped, turned around and established eye contact with the chaser across the fence (~1 s). The chaser raised their hands to the mouth and shouted (~2 s). Next, the chasee continued towards the box at the tunnel exit (~1 s). The lid of the box opened (accompanied by a clap sound) and the chasee jumped into it - after which the lid of the box closed, again accompanied by a clap sound (~1 s). Then, the chaser walked towards the tunnel entrance (~2 s) and transitted through the tunnel. While they were walking through the tunnel, footsteps sounded (~4 s - anticipatory period). After leaving the tunnel (~2 s), the chaser approached the box in which the chasee was hiding and knocks on it (~2 s). Then, the chasee

jumped out of the box (with a box opening clap sound) and the chaser and chasee did a high five (~4 s).

Test Trials

Test trials started with the same chasing sequence as in the familiarization trials. After doing a high five, chaser and chasee took their positions in front of the tunnel entrance. Next, the chasee made eye contact with the chaser, left for the tunnel and entered it. From this point onwards, the events depended on the condition:

In the *ignorance* condition, after the chasee entered the tunnel (~12 s after start), the chaser exited through the door in the wall in the back (~4 s). The back of the chaser remained visible. While the chaser was away (for ~8 s), the chasee walked through the tunnel (~4 s) and left through one of the exits (left or right, counterbalanced; ~2 s) and jumped into the respective box (~1 s). After approximately one second, while the chaser was still away, the chasee left this box A and tiptoed to the other box (~4 s). The chasee then jumped into box B and the lid closed (~1 s). In contrast to the familiarization trials, the chasee and the boxes made no sounds and no chime was played. After the hiding event had finished, the chaser returned through the door in the wall (~3 s) and entered the tunnel (~2 s). While the chaser was in the tunnel, footsteps were heard (~4 s). The video ended before the chaser exited the tunnel.

In the *knowledge* condition, the chaser remained on the scene in the upper section and positioned themselves centrally in front of the tunnel entrance (~2 s). Following the same sequence as in the ignorance condition, the chasee walked through the tunnel (~2 s), left it through one of the exits (left or right, counterbalanced) (~2 s) and hid in the respective box (~1 s). Next, in order to match the events of the ignorance condition, the chaser walked towards the door in the wall (~3 s) and disappeared for approximately 1 second. Subsequently, they returned to the initial position in front of the tunnel entrance (~3 s). In the meantime, the chasee did not move, so that

the chaser did not miss any events while they were gone. Once the chaser returned, they observed the chasee jumping out of the first box (~1 s) and tiptoeing to the second box (~4 s). Finally, the chasee jumped into the second box and the lid closed (~1 s). Like in the ignorance condition, the chasee and the boxes made no sound and no chime was played. The chaser entered the tunnel (~2 s) and footsteps sounded (~4 s). Like in the ignorance condition, the video ended before the chaser exited the tunnel.

Trial randomization

The four combinations in familiarization were the following: started from the right and ended up in right box (RR); started from the right and ended up in left box (RL); started from the left and ended up in right box (LR); started from the left and ended up in left box (LL). The presentation of the familiarization trials was counterbalanced in two pseudo-randomized orders (familiarization order A: Fam_LR, Fam_LL, Fam_RL, Fam_RR; familiarization order B: Fam_RL, Fam_RR, Fam_LR, Fam_LL). As with the familiarization trials, there were four different parallel versions of the test trial for the *knowledge* and the *ignorance* condition, differing in the starting location of the chasee and the box the chasee ended up (Know_RR, Know_RL, Know_LR, Know_LL; Ig_RR, Ig_RL, Ig_LR, Ig_LL). Supplementary Table S2 lists the combinations that were tested. Each lab signed up for one or two trial bins (16 trial combinations per bin) for each tested age group.

Table S2*Counterbalancing orders of parallel trial versions*

Trial bin	Trial order	Familiarization order	First (critical) test trial version	Second test trial version
1	1	A	KNOW_RR	IG_RR_ExitLeft_CON
	2	B	KNOW_RL	IG_RR_ExitLeft_CON
	3	B	KNOW_LR	IG_RR_ExitRight_INC
	4	A	KNOW_LL	IG_RR_ExitRight_INC
	5	B	KNOW_RR	IG_RL_ExitRight_CON
	6	A	KNOW_RL	IG_RL_ExitRight_CON
	7	A	KNOW_LR	IG_RL_ExitLeft_INC
	8	B	KNOW_LL	IG_RL_ExitLeft_INC
	9	A	IG_RR	KNOW_LR_ExitLeft_CON
	10	B	IG_RL	KNOW_LR_ExitLeft_CON
	11	B	IG_LR	KNOW_LR_ExitRight_INC
	12	A	IG_LL	KNOW_LR_ExitRight_INC
	13	B	IG_RR	KNOW_LL_ExitRight_CON
	14	A	IG_RL	KNOW_LL_ExitRight_CON
	15	A	IG_LR	KNOW_LL_ExitLeft_INC
	16	B	IG_LL	KNOW_LL_ExitLeft_INC
2	17	A	IG_RR	KNOW_RR_ExitRight_INC
	18	B	IG_RL	KNOW_RR_ExitRight_INC
	19	B	IG_LR	KNOW_RR_ExitLeft_CON
	20	A	IG_LL	KNOW_RR_ExitLeft_CON
	21	B	IG_RR	KNOW_RL_ExitLeft_INC
	22	A	IG_RL	KNOW_RL_ExitLeft_INC
	23	A	IG_LR	KNOW_RL_ExitRight_CON
	24	B	IG_LL	KNOW_RL_ExitRight_CON
	25	A	KNOW_RR	IG_LR_ExitRight_INC
	26	B	KNOW_RL	IG_LR_ExitRight_INC
	27	B	KNOW_LR	IG_LR_ExitLeft_CON
	28	A	KNOW_LL	IG_LR_ExitLeft_CON
	29	B	KNOW_RR	IG_LL_ExitLeft_INC
	30	A	KNOW_RL	IG_LL_ExitLeft_INC
	31	A	KNOW_LR	IG_LL_ExitRight_CON
	32	B	KNOW_LL	IG_LL_ExitRight_CON
3	33	A	KNOW_RR	IG_RR_ExitRight_INC
	34	B	KNOW_RL	IG_RR_ExitRight_INC
	35	B	KNOW_LR	IG_RR_ExitLeft_CON
	36	A	KNOW_LL	IG_RR_ExitLeft_CON
	37	B	KNOW_RR	IG_RL_ExitLeft_INC
	38	A	KNOW_RL	IG_RL_ExitLeft_INC
	39	A	KNOW_LR	IG_RL_ExitRight_CON
	40	B	KNOW_LL	IG_RL_ExitRight_CON
	41	A	IG_RR	KNOW_LR_ExitRight_INC
	42	B	IG_RL	KNOW_LR_ExitRight_INC
	43	B	IG_LR	KNOW_LR_ExitLeft_CON

Trial bin	Trial order	Familiarization order	First (critical) test trial version	Second test trial version
4	44	A	IG_LL	KNOW_LR_ExitLeft_CON
	45	B	IG_RR	KNOW_LL_ExitLeft_INC
	46	A	IG_RL	KNOW_LL_ExitLeft_INC
	47	A	IG_LR	KNOW_LL_ExitRight_CON
	48	B	IG_LL	KNOW_LL_ExitRight_CON
	49	A	IG_RR	KNOW_RR_ExitLeft_CON
	50	B	IG_RL	KNOW_RR_ExitLeft_CON
	51	B	IG_LR	KNOW_RR_ExitRight_INC
	52	A	IG_LL	KNOW_RR_ExitRight_INC
	53	B	IG_RR	KNOW_RL_ExitRight_CON
	54	A	IG_RL	KNOW_RL_ExitRight_CON
	55	A	IG_LR	KNOW_RL_ExitLeft_INC
	56	B	IG_LL	KNOW_RL_ExitLeft_INC
	57	A	KNOW_RR	IG_LR_ExitLeft_CON
	58	B	KNOW_RL	IG_LR_ExitLeft_CON
	59	B	KNOW_LR	IG_LR_ExitRight_INC
	60	A	KNOW_LL	IG_LR_ExitRight_INC
	61	B	KNOW_RR	IG_LL_ExitRight_CON
	62	A	KNOW_RL	IG_LL_ExitRight_CON
	63	A	KNOW_LR	IG_LL_ExitLeft_INC
	64	B	KNOW_LL	IG_LL_ExitLeft_INC

General Lab Practices

Training of Research Assistants

Each participating lab was responsible for maintaining the highest possible experimental standards, providing training practices for all experimenters and research assistants, and following detailed, written instructions to achieve uniformity and minimize variation across labs. Individual labs documented which experimenter(s) and research assistant(s) tested each participant. A questionnaire served to record and compare training practices. Greeting practices and instructions given to the participant/caregiver were marked down and standardized.

Reporting of Technology Mishaps and Participant/Caregiver Behavior

All labs were required to report anomalies, technical issues, concerns, and general comments on the protocol sheet. For toddler samples, concerns and general comments comprised

the following: crying, fussiness, weariness, caregiver intervening (verbal or non-verbal, e.g., pointing), affecting or disrupting participation and/or looking behavior. Technical issues included problems that hindered, paused, or stopped the stimulus presentation and/or eye-tracking recording.

Participant exclusion

Of the initial sample (toddlers: $N = 809$, adults: $N = 805$), participants were excluded from the main confirmatory analyses if: They did not complete the full experiment (toddlers: $n = 26$, 3.21%; adults: $n = 0$, 0%), participants' caregivers interfered with the procedure, e.g., by pointing at stimuli or talking to their toddler (toddlers: $n = 11$, 1.36%; adults: $n = 0$, 0%), the experimenter made an error during testing that was relevant to the procedure (toddlers: $n = 11$, 1.36%; adults: $n = 6$, 0.75%), technical problems occurred, e.g., data not saved, unable to calibrate eye-tracker, eye-tracker lost signal, data loss due to computer failure, computer crashed during recording (toddlers: $n = 69$, 8.53%; adults: $n = 61$, 7.58%). The individual labs determined whether and to which extent participant exclusion criteria 1-4 applied and added this information to the participant protocol sheet they provide. This set of exclusions left a total of 703 toddlers and 736 adults whose data will be analyzed. Of these, participants were excluded sequentially if: 5. Their data were excluded due to missingness (see Preprocessing section) from more than one familiarization trial (toddlers: $n = 112$, 13.84%; adults: $n = 24$, 2.98%), 6. Their data from the first (critical) test trial were excluded due to missingness (toddlers: $n = 54$, 6.67%; adults: $n = 9$, 1.12%). If multiple reasons for exclusion were applicable to a participant, the criteria were assigned in the order above.

S4. Further Supplemental Information: Analysis

Secondary analyses with less informative priors

To evaluate the sensitivity of our findings to prior assumptions, we performed two Bayesian analyses for each phase (familiarization and test) and age cohort (toddlers and adults) using the dependent variable of first look. The first analysis used weakly informative priors, reflecting modest assumptions about effect sizes and variability, as presented in the main manuscript. The second analysis employed less-informative priors, with fixed effect coefficients modeled using a normal prior ($N(0,3)$), and random effect standard deviations using ($N(0,0.5)$), allowing for greater flexibility in parameter estimation. For both prior types, we retained the same model structures used in the main analysis. In the familiarization phase, we fitted Bayesian mixed-effects models predicting target looks from trial number (1–4), including random intercepts and slopes for trial number by lab and participant. Trial number was coded so that the final familiarization trial (trial 4) was treated as the intercept, providing an estimate of anticipatory looking immediately before the test phase. The full model was compared to a simpler null model that excluded the intercept term. In the test phase, we modeled target looks as a function of condition (knowledge vs. ignorance, coded as $-0.5/0.5$), with random intercepts for lab. Here, the full model was compared to a null model that excluded the fixed effect of condition.

As shown in Table S3, results in the familiarization phase were highly consistent across both prior types and age cohorts. Bayes Factors exceeded 1000 in all cases, indicating strong evidence for the full model over the simpler null model without the intercept term, and parameter estimates differed only minimally between prior specifications. In the test phase, results were more sensitive to prior choice, particularly for toddlers. While parameter estimates remained

similar across priors, the Bayes Factor comparing the full model with the simpler null model without the fixed effect of condition decreased from 2.4 under the weakly informative prior to 1 under the less informative prior, suggesting that under the weakly informative prior, no conclusive inference could be drawn in support of - or against - the full model. In contrast, the adult test phase results remained robust across priors, with Bayes Factors consistently above 1000 and stable parameter estimates. These findings indicate that parameter estimates were generally stable across priors, Bayes Factors were more sensitive to prior specification when the evidence for the model was weaker, particularly in the toddler age cohort.

Table S3*Summary of Bayesian Results Across Phases and Priors*

Age cohort	Phase	Prior type	BF	Estimates
Toddlers	Familiarization	Weakly Informative	>1000	Intercept: 0.44 Main Effect: -0.22
		Less Informative	>1000	Intercept: 0.45 Main Effect: -0.23
Adults		Weakly Informative	>1000	Intercept: 1.03 Main Effect: 0.38
		Less Informative	>1000	Intercept: 1.06 Main Effect: 0.40
Toddlers	Test	Weakly Informative	2.4	Intercept: 0.53 Main Effect: 0.53 Interaction: -0.13
		Less Informative	1	Intercept: 0.57 Main Effect: 0.56 Interaction: -0.14
Adults		Weakly Informative	>1000	Intercept: 0.05 Main Effect: -0.89
		Less Informative	>1000	Intercept: 0.06 Main Effect: -0.96

Note. BFs reported for the familiarization phase are model comparisons of the full model with the simpler null model without the intercept. BFs reported for the test phase are model comparisons of the full model with the simpler model without the fixed effect of condition.

Data collection type: in-lab vs. web-based

In analyses introducing model terms for certain measurement characteristics, we quantified potential variability between in-lab and web-based data acquisition methods (cf., ManyBabies Consortium, 2020). Although our sample of participants tested with online sources was less than contributions of at least 32 participants), we conducted a separate analysis with a model term for online participants that estimated whether condition effects are different in this population.

In particular, Bayesian mixed-effects models were used to evaluate the effects of condition, method, and their interaction on proportional DLS. For toddlers, the effect of method was small and uncertain, with the credible interval including zero, indicating no clear effect of method. The interaction between condition and method was minimal and also uncertain, suggesting no strong evidence that the effect of condition varied by method. The estimated Bayes factor comparing the full model to the null model was approximately $BF = 0.7$, which indicates that the data slightly favored the null model over the full model. This suggests that the predictors included in the full model did not substantially improve the explanation of the observed data compared to the null model.

For adults, the main effect of method was slightly negative but uncertain, suggesting that the method had little to no clear effect on the outcome. The interaction between condition and method was negative but with a wide credible interval crossing zero, indicating uncertainty about whether the effect of condition varied by method. The estimated Bayes factor in favor of the full

model over the null model was $BF = 2.9$. This Bayes factor indicates that the evidence in favor of the model was modest but not strong. While the model was more likely than the null model to explain the observed data, the support is relatively weak, suggesting that the predictors in the model provided only a small improvement in explaining the data compared to the null model.

In sum, the analysis suggests that the method used (web-based vs. in-lab) did not have a strong impact on AL, as the effect of method and its interaction with condition were small and uncertain. Additionally, the results should be interpreted with caution due to the relatively small sample size for web-based data compared to in-lab data collection, which may limit the robustness of the findings.

AL as a function of differential fixation times of bear and mouse during location change of the mouse

In order to examine the effect of condition and the difference in looking times for mouse and bear during location change of the mouse on anticipatory looking, we fitted a Bayesian mixed-effects model for both age cohorts separately (see Figure S5 and Table S4). The dependent variable was the proportion DLS. The fixed effects included the main effects of condition, the difference in fixation times of mouse and bear, and their interaction. We also included random intercepts and slopes for differences in fixation times of mouse and bear within each participant and within each lab, allowing us to account for the hierarchical structure of the data and potential variability between labs and participants.

Due to limitations with bridge sampling, Bayes factors were computed using the Savage-Dickey density ratio method. These analyses provided moderate evidence against including both the fixed effect of difference in fixation times of mouse and bear, $BF = 0.3$, and its interaction with condition, $BF = 0.2$. These results suggest that the difference in looking time to mouse

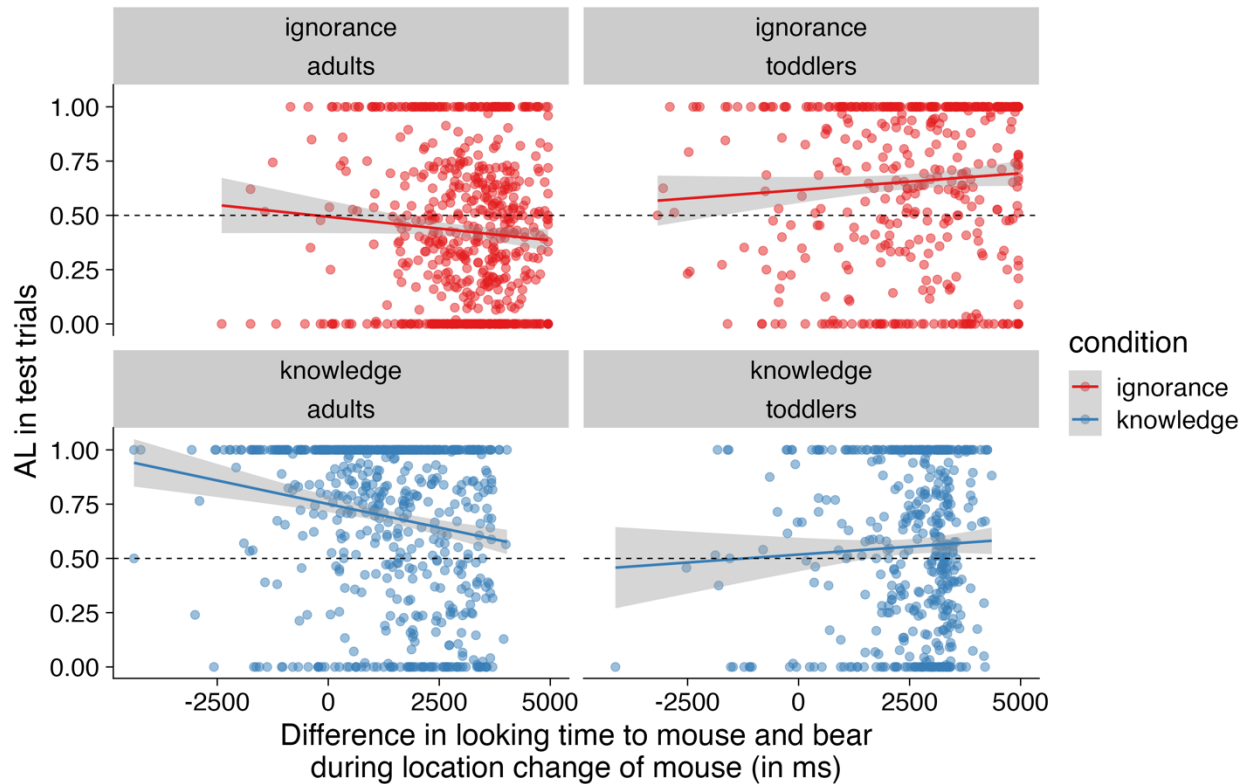
versus bear, either alone or in interaction with condition, did not meaningfully account for variance in toddlers' proportional DLS.

To assess whether proportional DLS in adults was predicted by condition, the difference in looking time between the mouse and bear, and their interaction, we fit a Bayesian mixed-effects model. There was a strong negative main effect of condition, indicating lower proportional DLS in the ignorance condition compared to the knowledge condition (Estimate = -0.21 , 95% CI [-0.29 , -0.13]). We also observed a negative main effect of mouse-bear looking difference, with greater looking toward the bear (relative to the mouse) associated with higher proportion DLS (Estimate = -0.04 , 95% CI [-0.05 , -0.02]). The interaction between condition and looking difference was close to zero and uncertain (Estimate = 0.01 , 95% CI [-0.02 , 0.04]).

Bayes factors computed using the Savage-Dickey method provided extreme evidence in favor of including the fixed effect of the difference in looking time between the mouse and bear, $BF > 1000$, and moderate evidence against including the interaction term, $BF = 0.2$. These results suggest that both condition and mouse-versus-bear looking independently predict proportional DLS in adults, but their interaction does not appear to contribute meaningfully.

Figure S6

AL as a function of the difference in looking time to mouse and bear during location change of mouse (in ms) for each age cohort and each condition



Note. Error bars represent 95% CIs.

Influence of target location from previous trials

Test phase: Influence of target location from last familiarization on DLS in first test trial

We examined whether proportion DLS in the first test trial was influenced by condition, the location of the target in the preceding familiarization trial (same vs. different), and their interaction. Separate Bayesian mixed-effects models were fitted for toddlers and adults, including random intercepts and slopes for condition by lab.

For toddlers, we found a significant effect of condition, with higher proportion DLS in the ignorance condition compared to the knowledge condition. The effect of location and the interaction between condition and location were not significant (see Table S5). A model comparison testing the inclusion of the location effect yielded a Bayes factor of $BF = 0.9$,

indicating anecdotal evidence against its contribution. A Bayes factor comparison indicated anecdotal evidence against the interaction model, with $BF = 0.6$.

For adults, results similarly revealed a significant effect of condition, with greater proportion DLS in the knowledge condition. The main effect of location and the interaction were not significant. The model comparison testing the main effect of location yielded a Bayes factor of $BF = 0.3$, suggesting moderate evidence against its inclusion. A Bayes factor comparison suggested anecdotal evidence against the inclusion of the interaction term, with $BF = 0.8$.

Familiarization phase: Influence of target location from previous familiarization trials on proportion DLS

We also assessed whether proportion DLS during the familiarization phase was influenced by the target's location in the previous trial, trial number, and their interaction. Separate Bayesian mixed-effects models were fitted for toddlers and adults, with random intercepts for participants and labs.

For toddlers, there was a significant negative effect of previous trial location (see Table S5), indicating that toddlers looked more at the target when it had been on the same side as in the previous trial. The effect of trial number and the interaction were not significant. The model comparison testing the contribution of the location effect alone yielded a Bayes Factor of $BF > 1000$, indicating extreme evidence in favor of including previous trial location. Furthermore, a Bayes factor analysis provided anecdotal evidence in favor of the full model compared to the null model without the interaction of these predictors, $BF = 1.1$.

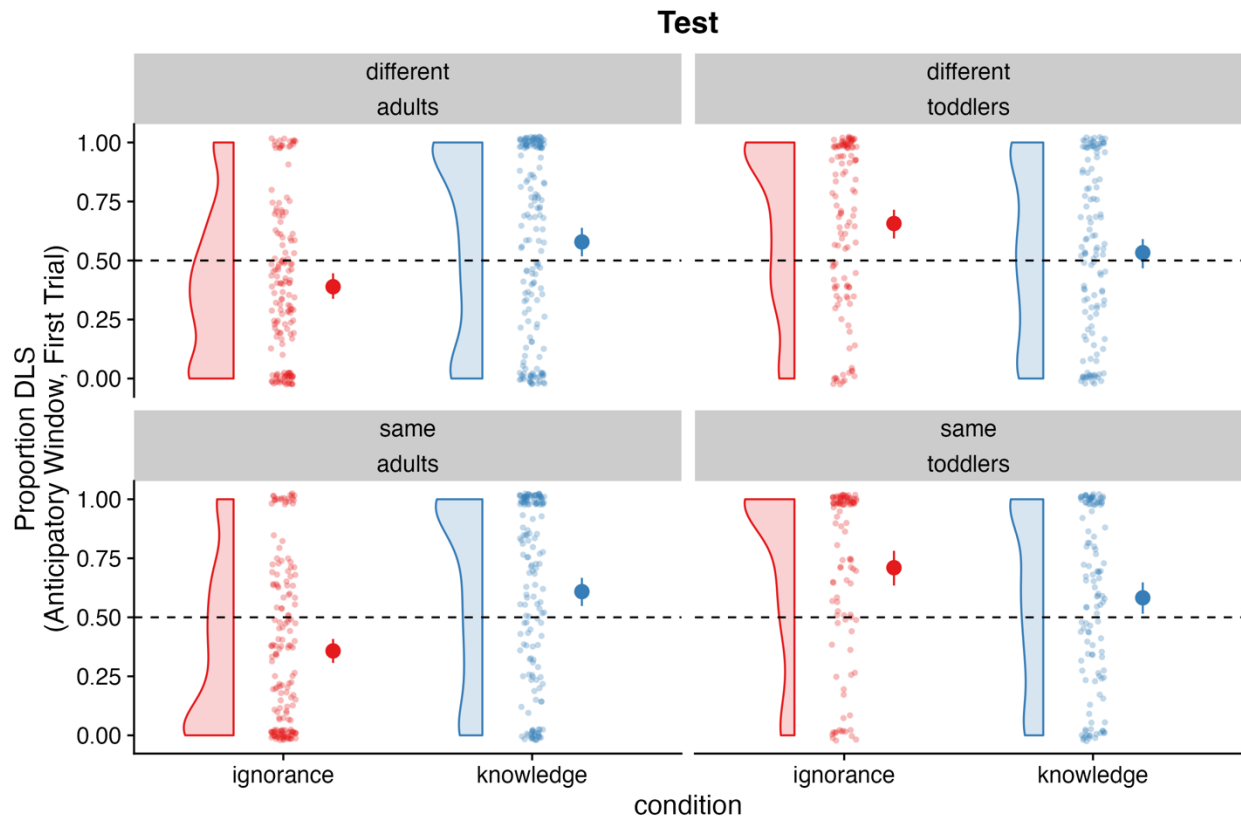
For adults, no predictor reached significance. The Bayes factor for including location was $BF = 1.3$, providing only anecdotal support for its contribution. The Bayes factor, $BF = 0.4$, further supports the conclusion that the model including the interaction of the predictors was not substantially better than the null model, providing anecdotal evidence for the null model.

Table S5*Results of the Bayesian mixed effects models for consistency of location*

model	term	β	SE	lower CrI95%	upper CrI95%	$\hat{R}$
test adults	Intercept	0.48	0.02	0.45	0.52	1.00
	Condition	-0.20	0.03	-0.26	-0.15	1.00
	Location	0.00	0.03	-0.05	0.05	1.00
	Condition:Location	0.05	0.05	-0.05	0.14	1.00
test toddlers	Intercept	0.62	0.02	0.59	0.66	1.00
	Condition	0.11	0.03	0.05	0.18	1.00
	Location	-0.04	0.03	-0.11	0.02	1.00
	Condition:Location	0.00	0.06	-0.11	0.11	1.00
fam adults	Intercept	0.19	0.04	0.12	0.26	1.00
	Location	0.09	0.06	-0.03	0.22	1.00
	Trial Number	0.02	0.03	-0.04	0.08	1.00
	Location:Trial Number	0.00	0.06	-0.12	0.12	1.00
fam toddlers	Intercept	0.15	0.03	0.09	0.22	1.00
	Location	-0.10	0.06	-0.22	0.01	1.00
	Trial Number	-0.03	0.03	-0.08	0.03	1.00
	Location:Trial Number	0.07	0.05	-0.04	0.17	1.00

Figure S7

Proportion DLS during the test phase for toddlers and adults split by condition (knowledge vs. ignorance) and consistency of location (same vs. different)



Note. Error bars represent 95% CIs.

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